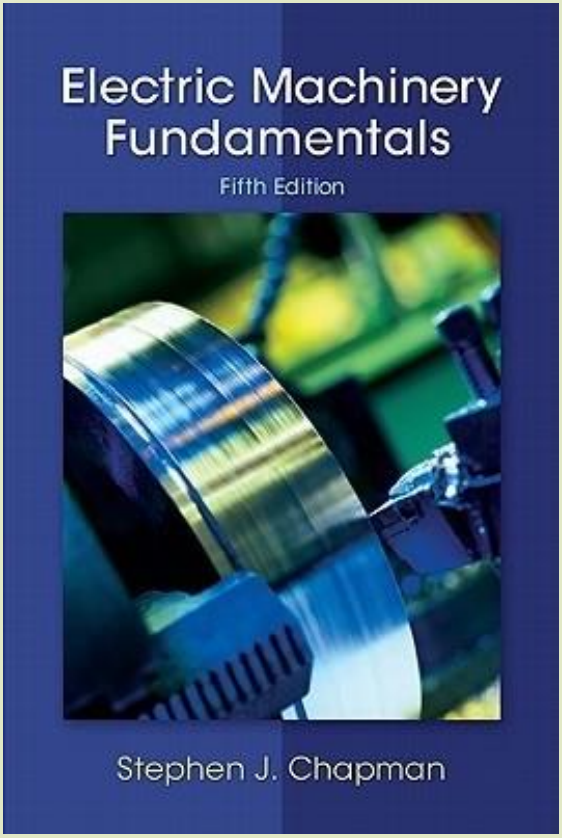
Chapter 1
Introduction to Machinery Principles

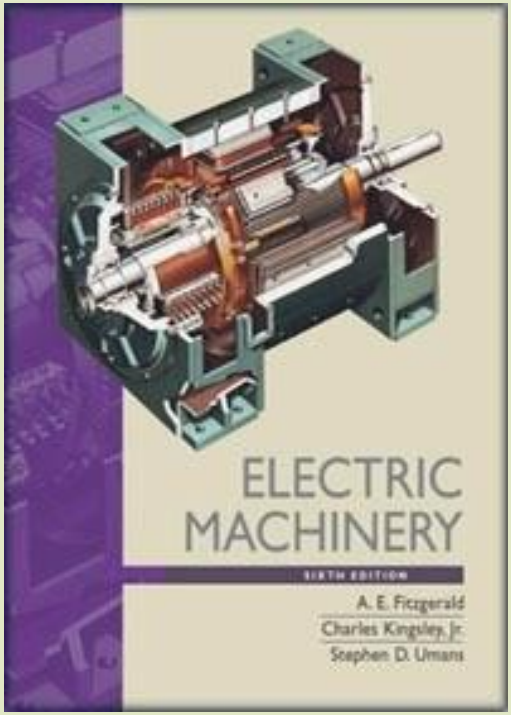
Text book and supplementary materials of this course

- Stephen J. Chapman
- 5th edition (Feb. 18, 2011)



Reference book

- A. E. Fitzgerald, *Electric Machinery*, McGraw-Hill, 6th edition (July 25, 2002)



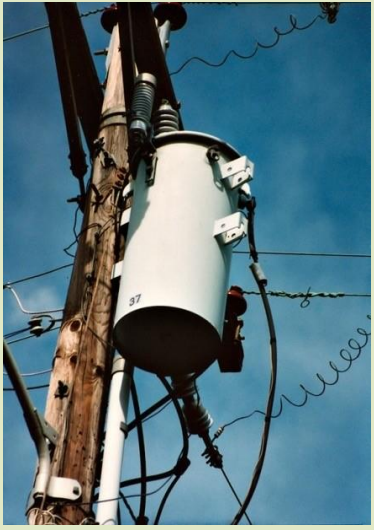
Electric Machinery Principles













Energy types

Energy is the ability to do work or cause a change. There are many types of energy, including:

Thermal energy - the energy of heat, which can be converted into mechanical energy or electrical energy through processes such as combustion, nuclear reactions, or thermoelectric conversion.

Electrical energy - the energy of moving electrons, which can be generated through processes such as electromechanical conversion or photoelectric conversion.

Energy types

Chemical energy - the energy stored in chemical bonds, which can be released through processes such as combustion or cellular respiration.

Light energy - the energy of electromagnetic waves, which can be converted into electrical energy through processes such as photovoltaic conversion.

Energy conversions

Energy can be converted from one form to another through various processes. Some examples of energy conversions are:

Chemical to electrical - in a battery, chemical energy is converted into electrical energy through a redox reaction.

Nuclear to thermal - in a nuclear power plant, nuclear energy is converted into thermal energy through nuclear fission, which is then used to produce steam and generate electricity.

Energy conversions

Thermal to mechanical - in a steam engine, thermal energy from the combustion of fuel is used to produce steam, which drives a piston to produce mechanical energy.

Electrical to thermal - in an electric heater, electrical energy is converted into thermal energy through the resistance of a wire.

Energy conversions

Mechanical to electrical - in a generator, mechanical energy from a turbine is used to turn a rotor in a magnetic field, generating electrical energy.

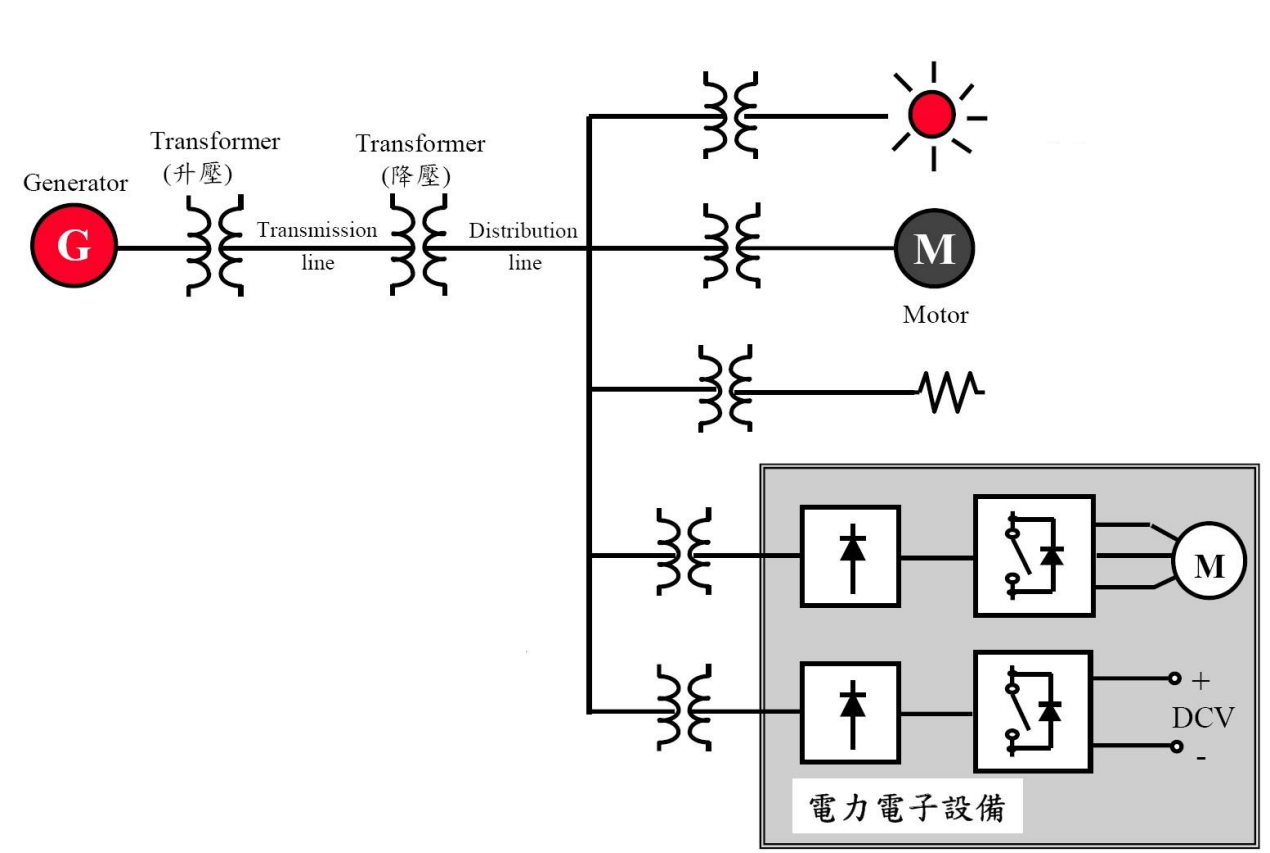
Light to electrical - in a solar cell, light energy is converted into electrical energy through the photoelectric effect.

Energy conversions

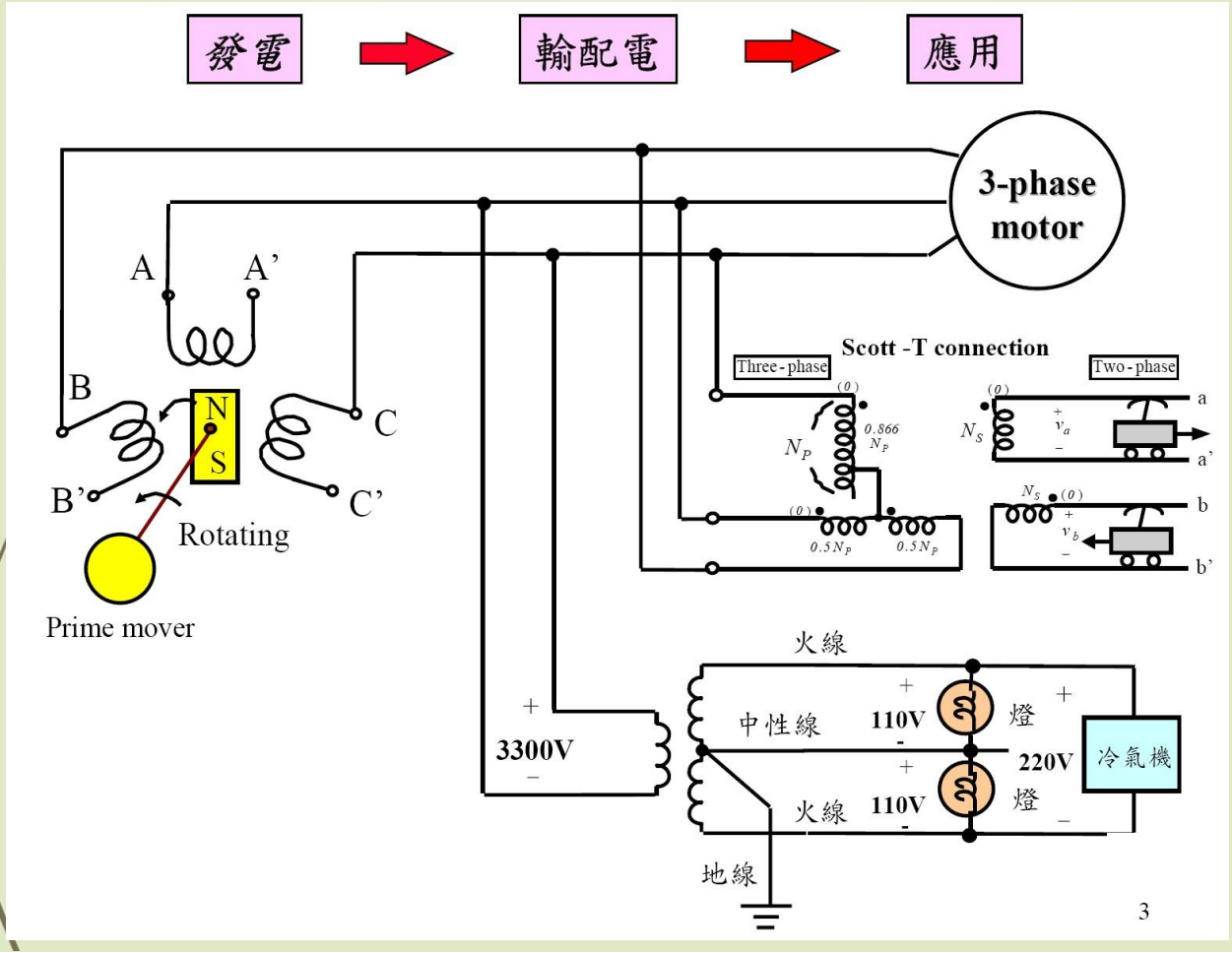
Energy conservation is important to reduce the environmental impact of energy use and to ensure a sustainable future.

It is also important to consider the efficiency of energy conversions, as some forms of energy conversion may result ins ignificant losses of energy.

Introduction to Electric Machinery Fundamentals



Introduction to Electric Machinery Fundamentals



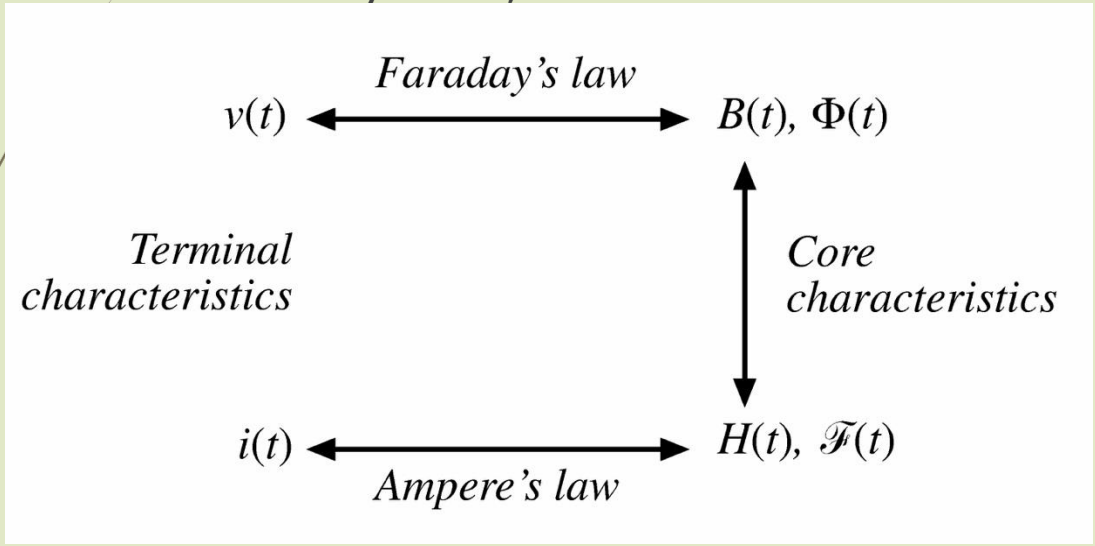
What to learn in this course

- **Energy Conversion** schemes are the key ideas introduced in this course
- Which types of energy conversion are concerned?
- Electric energy to electric energy
  - **Transformer**
- Electric energy to mechanical energy
  - **Motor**
- Mechanical energy to electric energy
  - **Generator**
- Magnetic energy is essential !

Course Outlines

Overview of relative electromagnetic theories

- Magnetic field: Ampere's law
- Magnetic flux: magnetic material, hysteresis characteristics
- Transformer: Faraday's law, Len's law



Course Outlines

Overview of Relative Electromagnetic Theories

- **Magnetic circuit:**
- **Motor/generator:** Induced voltage, induced force

Course Outlines
Transformer

- Ideal/non-ideal transformer
- Equivalent transformer circuit
- Voltage regulation, efficiency
- Determining transformer parameters
  - Open-circuit test
  - Short-circuit test

Course Outlines

Basic electric machine (motor/generator) theories

- **AC machine**
  - induction machine
  - synchronous machine
- **DC machine**
  - separated excited,
  - shunt excited,
  - series excited,
  - compound excited
- **How the motor rotates**
  - Torque/speed
- **How the generator to build output voltage**
  - Voltage/current

Course Outline

Induction (Asynchronous) Machine

- Induction motor (IM) – the most widely used ac motor in the world
  - Structure and operation theories of IM
  - Equivalent circuit of IM
  - Torque/speed characteristics
  - Basic motor control
- Induction generator (seldom used)
  - Output voltage control
  - Voltage/current characteristics

Course Outline
Synchronous machines

- Synchronous generator (SG) – the most widely used generator in the world
  - Structure and operation theories of SG
  - Equivalent circuit of SG
  - Voltage/current characteristics
  - Parallel operation
- Synchronous motor
  - Operation principles
  - Starting of synchronous motor
  - Torque/speed characteristics

History of Electric Machinery



DC generator, driven by steam engines

Waterwheel-driven DC generator installed in Appleton, Wisconsin

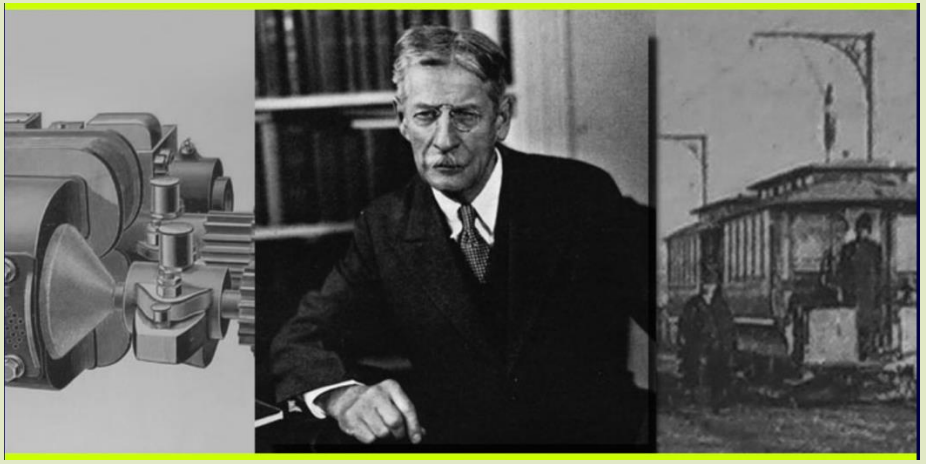
1882 Thomas A. Edison opens Pearl St. Station, NYC



History of Electric Machinery

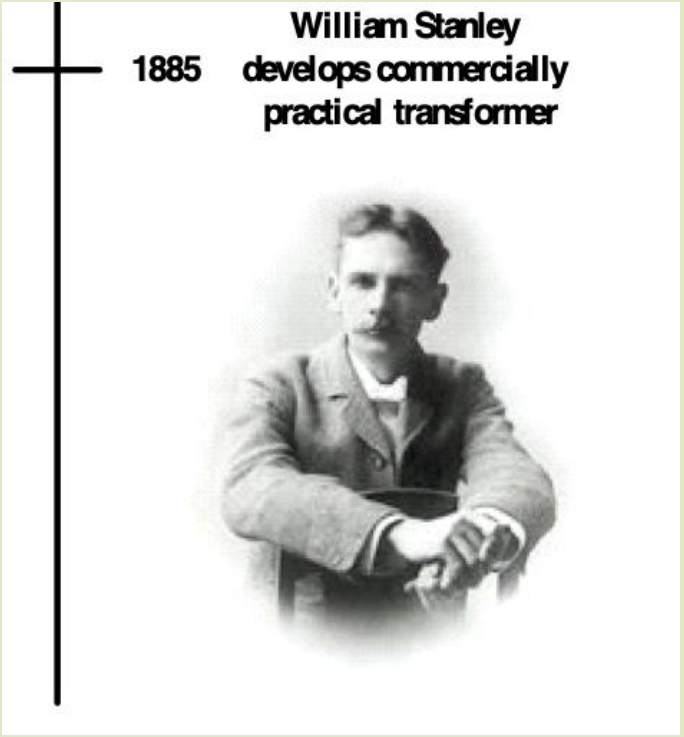


1884 Frank J. Sprague produces DC motor for Edison systems

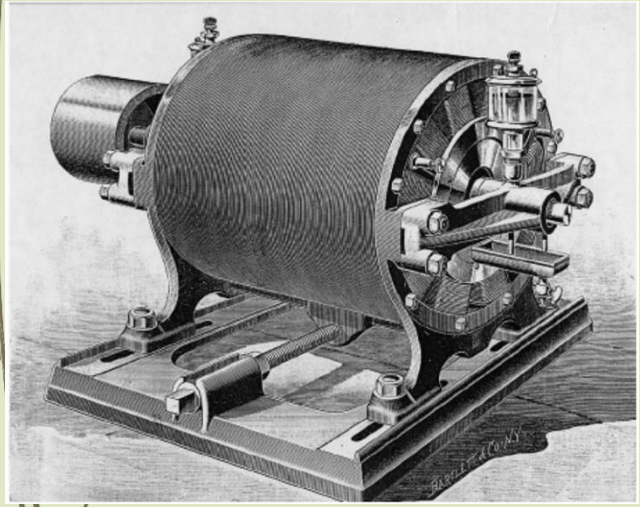


History of Electric Machinery



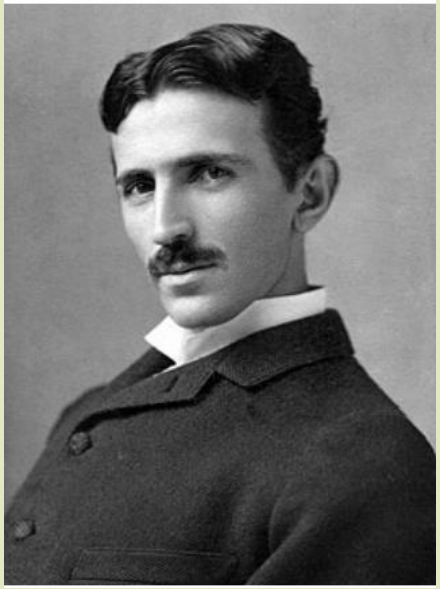


History of Electric Machinery



The TESLA AC Motor—1888.

1888 Nikola Tesla presents paper on two-phase ac induction and synchronous motors



History of Electric Machinery

The history of electrical machines dates back to the early 19th century when scientists began to experiment with the principles of electromagnetism.

In 1821, Michael Faraday discovered that electric current could produce a magnetic field, and

In 1824 he invented the first electric motor, which consisted of a wire hanging from a battery that could rotate around a magnet.

History of Electric Machinery

In the following years, other scientists and inventors made important contributions to the development of electrical machines.

In 1831, Joseph Henry built the first practical electromagnetic generator, which could produce an electric current from a rotating magnet.

In 1832, Hippolyte Pixii built the first alternating current (AC) generator, which used a commutator to convert the output of a rotating magnet into a pulsating DC current.

History of Electric Machinery

In the late 19th century, Nikola Tesla and George Westinghouse made significant contributions to the development of AC power systems.

Tesla invented the induction motor, which could operate on AC power, and Westinghouse built the first AC power system, which used transformers to distribute power over long distances.

During the 20th century, electrical machines became more sophisticated and efficient, thanks to advances in materials, manufacturing techniques, and control systems.

History of Electric Machinery

In the early years of the century, the DC motor was the dominant type of electrical machine.

But by the mid-20th century, AC motors had become more common, especially for industrial applications.

Today, electrical machines are used in a wide range of applications, from small appliances to large industrial machinery.

And continue to evolve as new technologies emerge, such as electric vehicles and renewable energy systems.

Today's development

- DC Machine
  - Motor
  - Generator
- Transformer
  - Single phase
  - Three phases
- AC Machine
  - Synchronous machine – motor, generator
  - Asynchronous machine (induction machine) – motor, generator

Today's development and future trends

- Micro-step stepping motor
- Permanent magnet synchronous motor (PMSM)
  - Brushless dc motor (BLDCM)
- Linear motor
- Reluctance motor
  - Synchronous reluctance
  - Switched reluctance
- Ultrasonic motor
- Bionic robotics
- MEMS motor

Course relation

- It is the fundamental course of the electrical engineering
- Future courses
  - Power electronics
  - Motor control
  - Electric motor drive
  - Power systems
  - Renewable energy
  - Electrical vehicle

Chapter 1 Introduction to machinery principles

1. Rotation motion, Newton's law and power relationships
2. The magnetic field
3. Faraday's law
4. Produce an induced force on a wire
5. Produce an induced voltage on a conductor
6. Linear dc machine examples
7. Real, reactive and apparatus power in AC circuits

Rotation motion, Newton's law and power relationships

- Clockwise (CW) and Counterclockwise (CCW)
  - CCW is assumed as the positive direction, CW is assumed as the negative direction.
- One dimension and rotation motion
  - Position and angular (meter) (degree or radian)
  - Speed and angular speed (meter/s) (degree/s or radian/s)

$$v = \frac{dr}{dt}$$

$$\omega = \frac{d\theta}{dt}$$

ω_m	angular velocity expressed in radians per second
f_m	angular velocity expressed in revolutions per second
n_m	angular velocity expressed in revolutions per minute

Rotation motion, Newton's law and power relationships

- Revolution and frequency relationship

$$n_m = 60 f_m$$

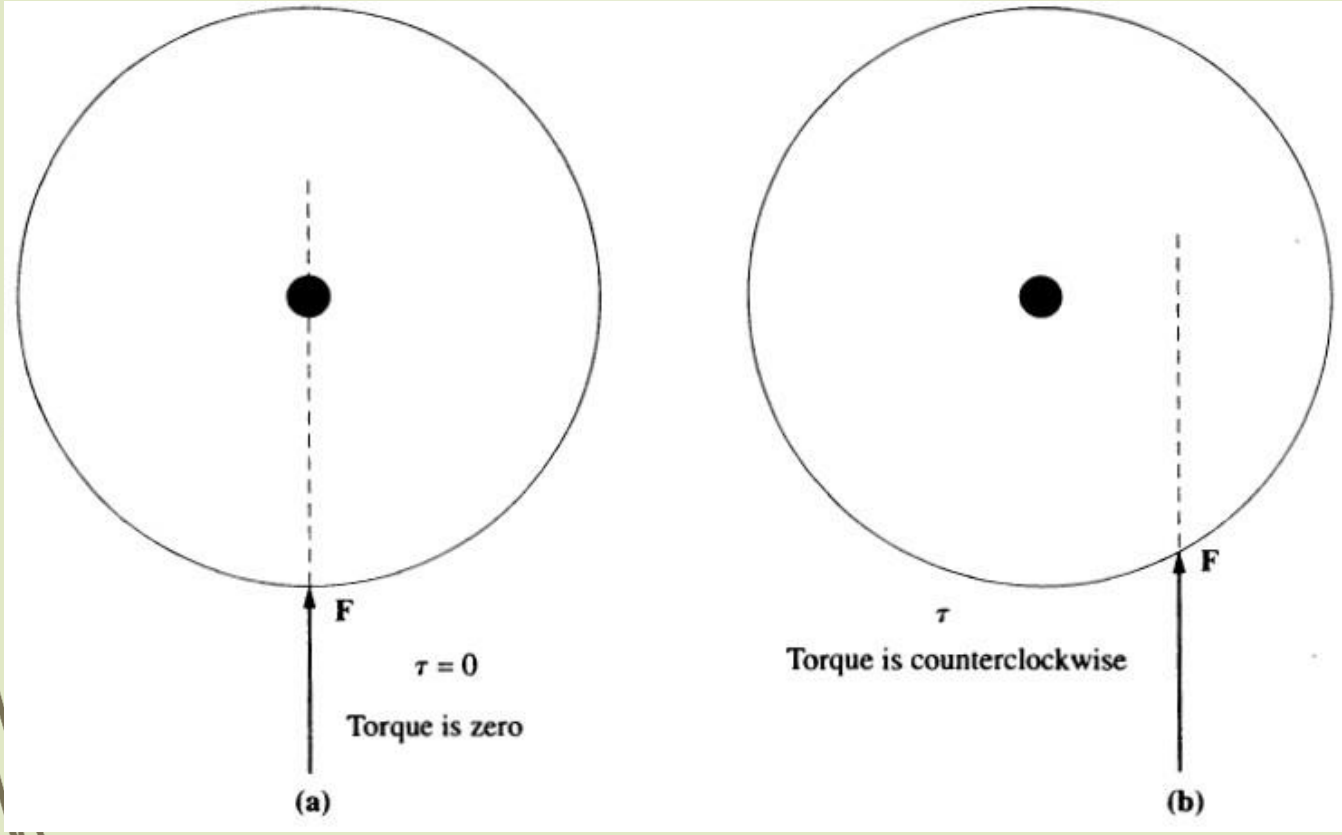
$$f_m = \frac{\omega_m}{2\pi}$$

- Acceleration and angular acceleration

$$a = \frac{dv}{dt}$$

$$\alpha = \frac{d\omega}{dt}$$

Torque



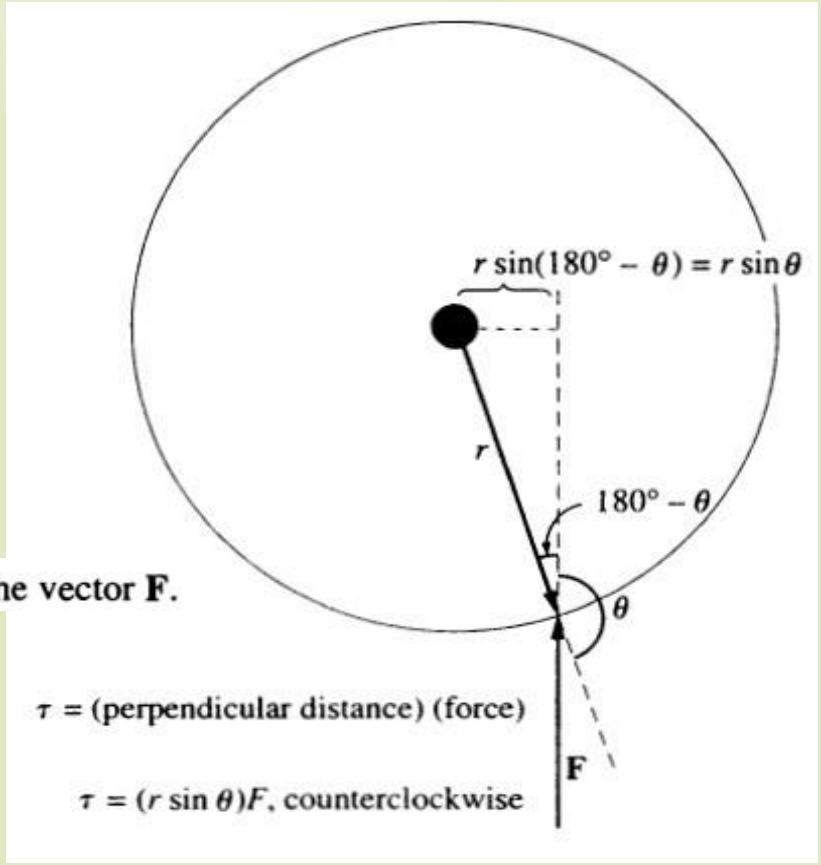
Torque

τ = (force applied)(perpendicular distance)

$= (F)\,(r \sin \theta)$

$= rF \sin \theta$

where θ is the angle between the vector **r** and the vector **F**.



Torque and Work

For linear motion, work is defined as the application of a *force* through a *distance*. In equation form,

$$W = \int F\,dr \tag{1–9}$$

where it is assumed that the force is collinear with the direction of motion. For the special case of a constant force applied collinearly with the direction of motion, this equation becomes just

$$W = Fr \tag{1–10}$$

The units of work are joules in SI and foot-pounds in the English system.

For rotational motion, work is the application of a *torque* through an *angle*. Here the equation for work is

$$W = \int \tau\,d\theta \tag{1–11}$$

and if the torque is constant,

$$W = \tau\theta \tag{1–12}$$

Newton's law of rotation

1. Force

$$F = ma$$

F = net force applied to an object
m = mass of the object
a = resulting acceleration

2. Torque

$$\tau = J\alpha$$

Power (rate of doing work)

Power is the rate of doing work, or the increase in work per unit time. The equation for power is

$$P = \frac{dW}{dt} \tag{1–13}$$

It is usually measured in joules per second (watts), but also can be measured in foot-pounds per second or in horsepower.

By this definition, and assuming that force is constant and collinear with the direction of motion, power is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(Fr) = F\left(\frac{dr}{dt}\right) = Fv \tag{1–14}$$

Similarly, assuming constant torque, power in rotational motion is given by

$$P = \frac{dW}{dt} = \frac{d}{dt}(\tau\theta) = \tau\left(\frac{d\theta}{dt}\right) = \tau\omega$$

$$P = \tau\omega \tag{1–15}$$

Equation (1–15) is very important in the study of electric machinery, because it can describe the mechanical power on the shaft of a motor or generator.

Relationships of linear Motion and Revolution

Linear Motion		Revolution	
r	*distance*	θ	*angular*
v	$v = \frac{dr}{dt}$	w	$w = \frac{d\theta}{dt}$
a	$a = \frac{dv}{dt}$	α	$\alpha = \frac{dw}{dt}$
f	*force*	τ	*torque*
W	$W = fr$	W	$W = \tau\theta$
P	$P = \frac{dW}{dt} = Fv$	P	$P = \frac{d}{dt}(\tau\theta) = \tau w$

Conversion between watts and horsepower

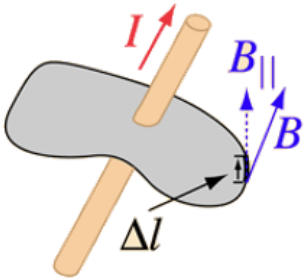
$$\mathbf{P(watts)= } \tau\ (Nm)\omega(\frac{rad}{s})$$

1hp = 746W = 0.746kW

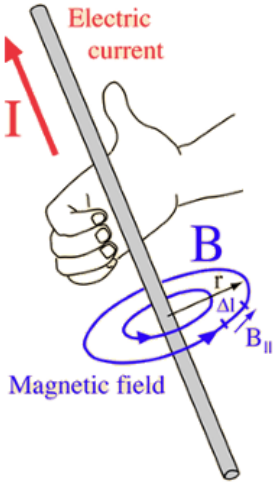
The magnetic field

1. A current-carrying wire produces a magnetic field in the area around it.
2. A time-changing magnetic field induces a voltage in a coil of wire if it passes through that coil. (This is the basis of *transformer action*.)
3. A current-carrying wire in the presence of a magnetic field has a force induced on it. (This is the basis of *motor action*.)
4. A moving wire in the presence of a magnetic field has a voltage induced in it. (This is the basis of *generator action*.)

1) A **current-carrying wire** produces a magnetic field in the area around it. **(Ampere's Law)**



$$\sum B_{||} \Delta l = \mu_0 I$$



Ampere's Law states that for any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop.

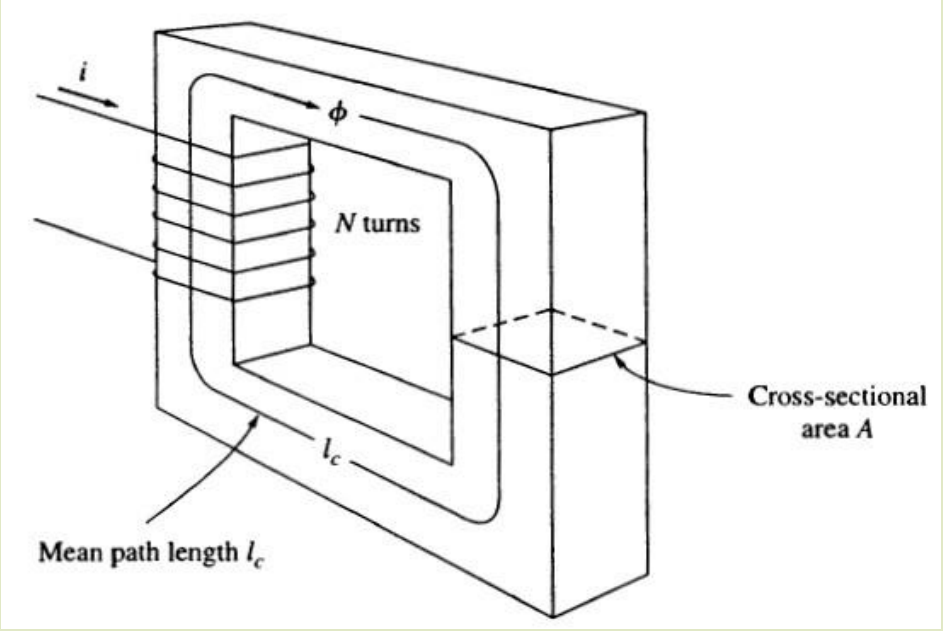
Producing a magnetic field
Ampere's law

1. The magnetic field is produced by Ampere's law
2. The core is a ferromagnetic material

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{net}}$$

$$Hl_c = Ni$$

$$H = \frac{Ni}{l_c}$$



From the magnetic field to magnetic flux density

1. When the magnetic field is applied on a ferromagnetic material, the magnetic flux density **B** will be produced

$$\mathbf{B} = \mu\mathbf{H}$$

H = magnetic field intensity
μ = magnetic *permeability* of material
B = resulting magnetic flux density produced

The permeability of free space is called μ_0, and its value is

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \qquad (1\text{–}22)$$

The permeability of any other material compared to the permeability of free space is called its *relative permeability:*

$$\mu_r = \frac{\mu}{\mu_0} \qquad (1\text{–}23)$$

Unit conversions

Units

Table 12.1. Units for magnetic quantities

quantity	*MKS*	*unrationalized cgs*	*conversions*
core material equation	$B = \mu_0 \mu_r H$	$B = \mu_r H$	
B	Tesla	Gauss	$1T = 10^4 G$
H	Ampere / meter	Oersted	$1A/m = 4\pi \cdot 10^{-3}$ Oe
Φ	Weber	Maxwell	$1Wb = 10^8$ Mx $1T = 1Wb / m^2$

Magnetic flux density and magnetic flux

1. Magnetic flux density

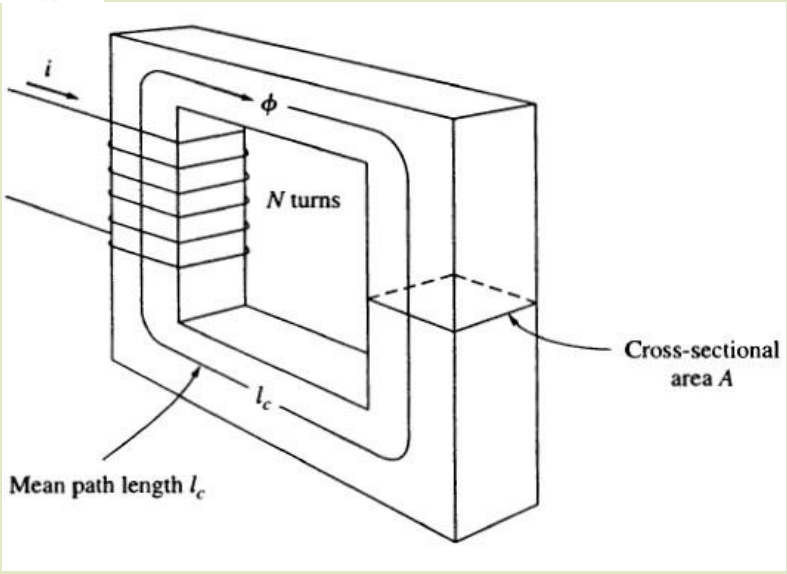
$$B = \mu H = \frac{\mu N i}{l_c}$$

2. Magnetic flux

$$\phi = \int_A \mathbf{B} \cdot d\mathbf{A}$$

$$\phi = BA$$

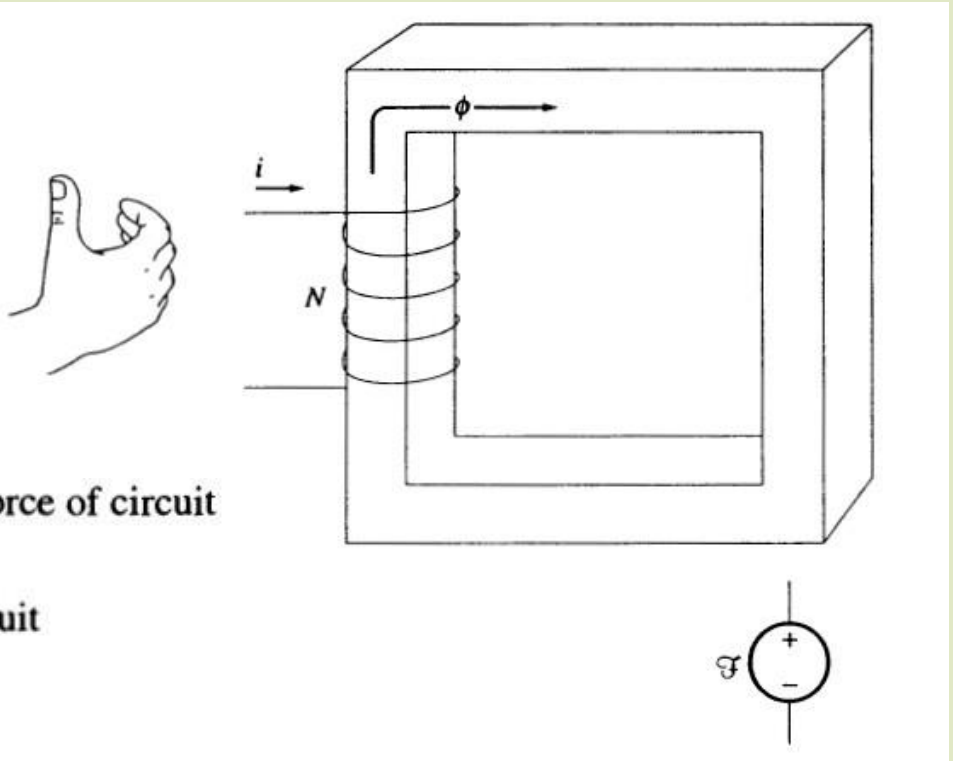
$$\phi = BA = \frac{\mu N i A}{l_c}$$



Magnetic Circuit – magnetomotive force

$\mathcal{F} = Ni$

$\mathcal{F} = \phi \mathcal{R}$



$\mathcal{F}$ = magnetomotive force of circuit

ϕ = flux of circuit

$\mathcal{R}$ = *reluctance* of circuit

Magnetic circuit

$$\phi = BA = \frac{\mu NiA}{l_c}$$

$$= Ni\left(\frac{\mu A}{l_c}\right)$$

$$\phi = \mathcal{F}\left(\frac{\mu A}{l_c}\right)$$

$$\mathcal{R} = \frac{l_c}{\mu A}$$

$$\mathcal{F} = \phi \mathcal{R}$$

Electric circuit and magnetic circuit

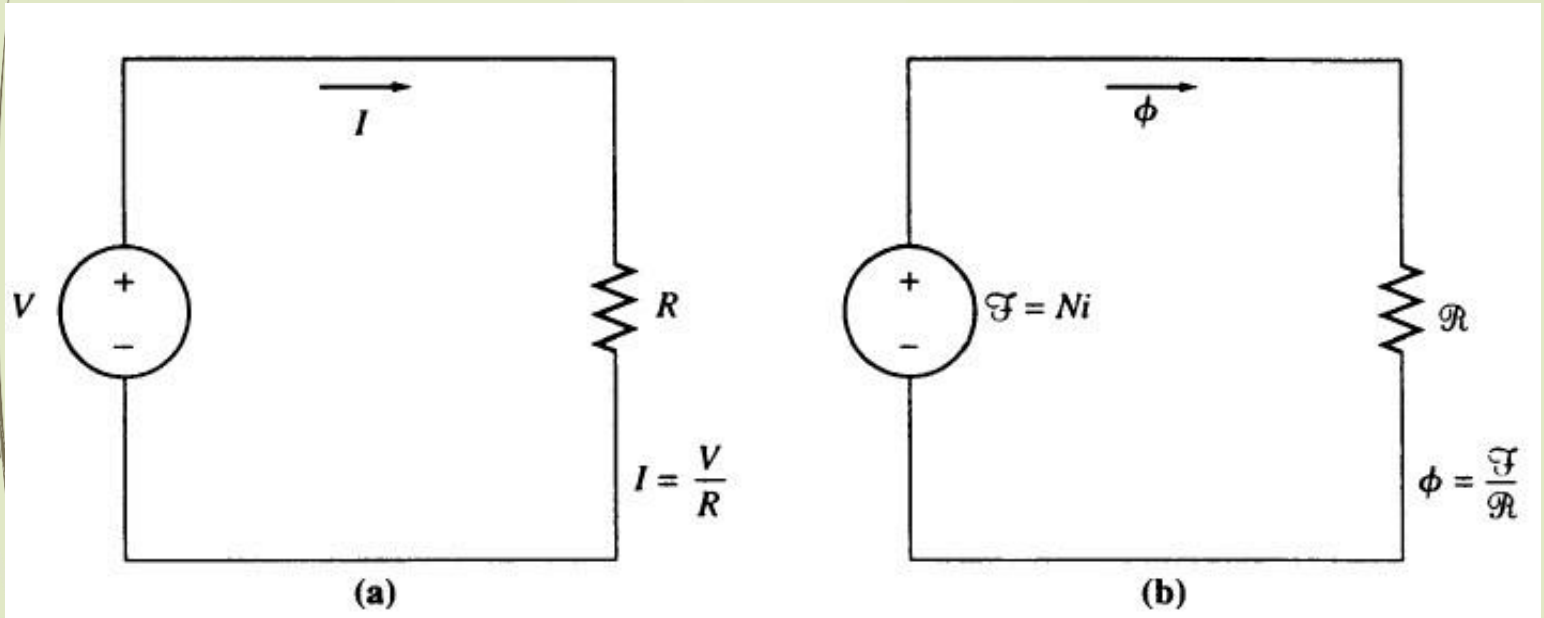


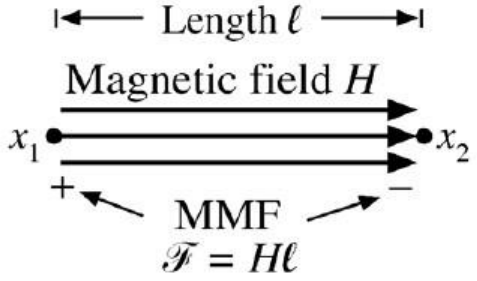
FIGURE 1–4
(a) A simple electric circuit. (b) The magnetic circuit analog to a transformer core.

Magnetic field H and magnetomotive force $\mathscr{F}$

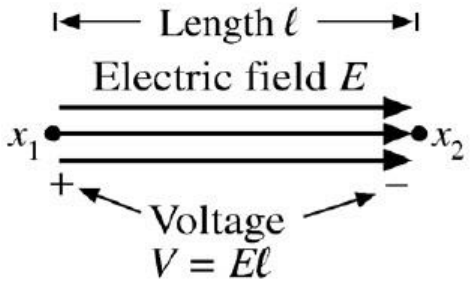
Magnetomotive force (MMF) $\mathscr{F}$ between points x_1 and x_2 is related to the magnetic field $\boldsymbol{H}$ according to

$$\mathscr{F} = \int_{x_1}^{x_2} \boldsymbol{H} \cdot d\boldsymbol{\ell}$$

Example: uniform magnetic field of magnitude H



Analogous to electric field of strength E, which induces voltage (EMF) V:



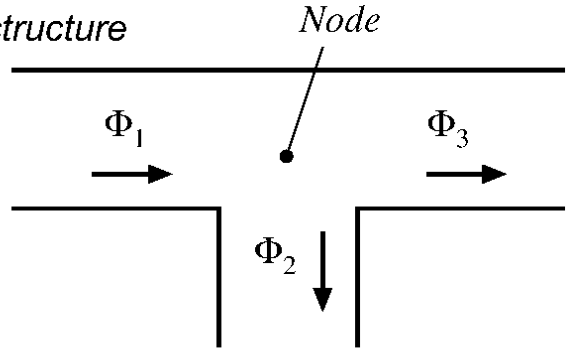
Magnetic analog of Kirchoff's current law

Divergence of $\boldsymbol{B} = 0$

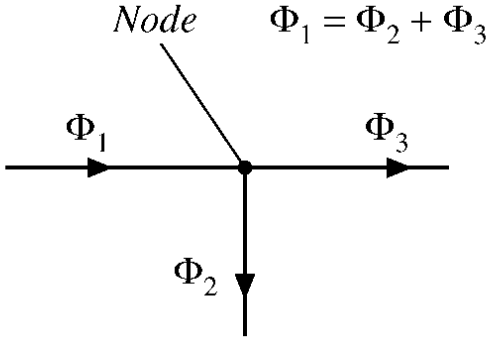
Flux lines are continuous and cannot end

Total flux entering a node must be zero

Physical structure



Magnetic circuit



Reluctance in magnetic circuit

1. Series connection

$$\mathcal{R}_{eq} = \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \cdots$$

2. Parallel connection

$$\frac{1}{\mathcal{R}_{eq}} = \frac{1}{\mathcal{R}_1} + \frac{1}{\mathcal{R}_2} + \frac{1}{\mathcal{R}_3} + \cdots$$

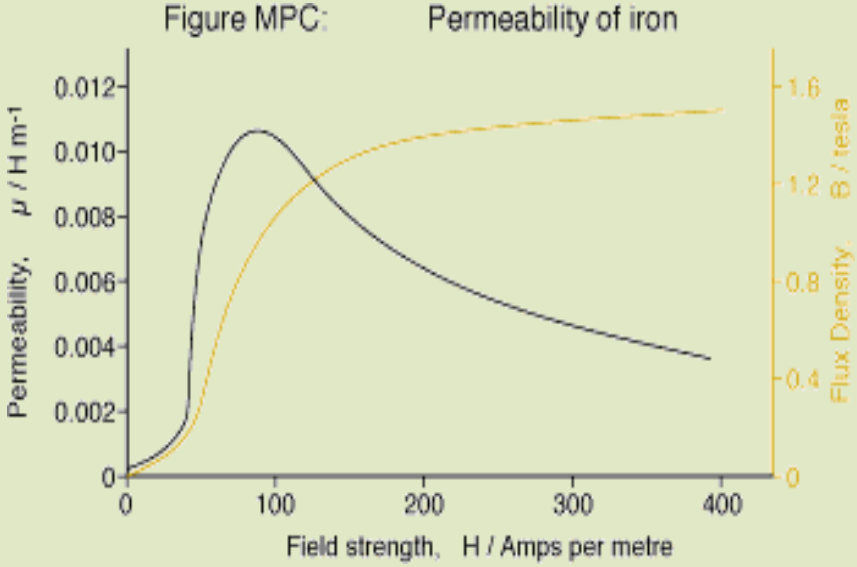
The errors in magnetic circuit computation

1. The magnetic circuit concept assumes that all flux is confined within a magnetic core. Unfortunately, this is not quite true. The permeability of a ferromagnetic core is 2000 to 6000 times that of air, but a small fraction of the flux escapes from the core into the surrounding low-permeability air. This flux outside the core is called *leakage flux*, and it plays a very important role in electric machine design.
2. The calculation of reluctance assumes a certain mean path length and cross-sectional area for the core. These assumptions are not really very good, especially at corners.
3. In ferromagnetic materials, the permeability varies with the amount of flux already in the material. This nonlinear effect is described in detail. It adds yet another source of error to magnetic circuit analysis, since the reluctances used in magnetic circuit calculations depend on the permeability of the material.
4. Airgap fringing effect

The errors in magnetic circuit computation (computation errors)

- Why are the calculations in a magnetic circuit **approximate** ? (*Accuracy is up to **5%***)

In ferromagnetic materials, the permeability **varies** with the amount of flux already in the material. Because of this nonlinearity, in practice, the reluctance values are **not constant**.



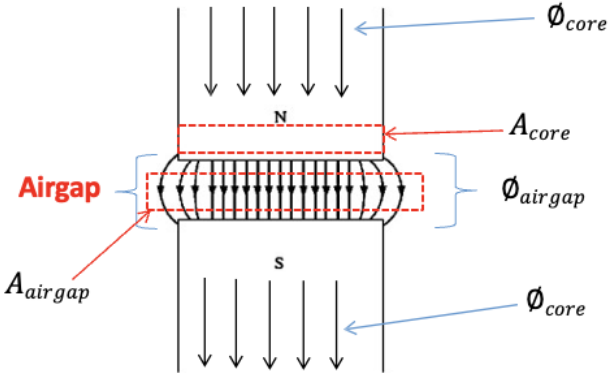
Source: http://info.ee.surrey.ac.uk/Workshop/advice/coils/mu/

Production of a Magnetic Field

- Why are the calculations in a magnetic circuit **approximate** ? (*Accuracy is up to **5%***)

If there are air gaps in the flux path in a core, the effective cross-sectional area of the air gap will be larger than the cross-sectional area of the iron core on either side.

The extra effective area is caused by the "fringing effect" of the magnetic field at the air gap.



$\emptyset_{core} = \emptyset_{airgap}$ → Think like series connected resistances have the **same current**

$A_{airgap} > A_{core}$

Since;

$$B = \frac{\phi}{A}$$

$B_{core} > B_{airgap}$ **(Fringing Effect)**

Example 1-1

Example 1–1. A ferromagnetic core is shown in Figure 1–7a. Three sides of this core are of uniform width, while the fourth side is somewhat thinner. The depth of the core (into the page) is 10 cm, and the other dimensions are shown in the figure. There is a 200-turn coil wrapped around the left side of the core. Assuming relative permeability μ_r of 2500, how much flux will be produced by a 1-A input current?

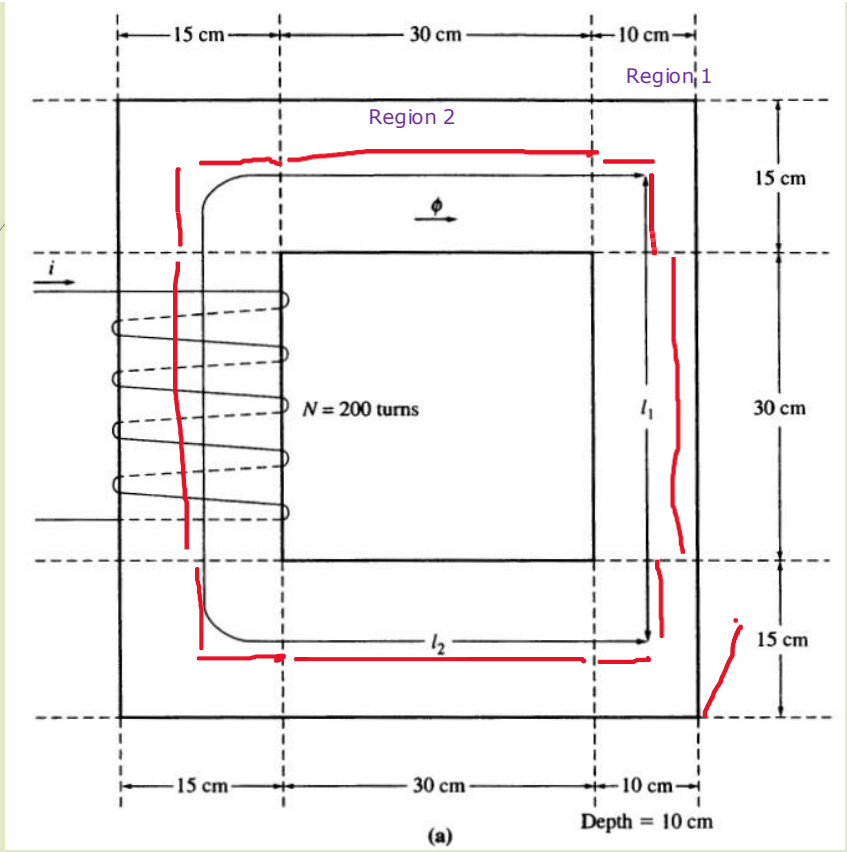


Fig. 1-7a

Example 1-1

The mean path length of region 1 is 45 cm, and the cross-sectional area is 10 × 10 cm = 100 cm². Therefore, the reluctance in the first region is

$$\mathcal{R}_1 = \frac{l_1}{\mu A_1} = \frac{l_1}{\mu_r \mu_0 A_1} \tag{1–32}$$

$$= \frac{0.45 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.01 \text{ m}^2)}$$

$$= 14{,}300 \text{ A} \cdot \text{turns/Wb}$$

The mean path length of region 2 is 130 cm, and the cross-sectional area is 15 × 10 cm = 150 cm². Therefore, the reluctance in the second region is

$$\mathcal{R}_2 = \frac{l_2}{\mu A_2} = \frac{l_2}{\mu_r \mu_0 A_2} \tag{1–32}$$

$$= \frac{1.3 \text{ m}}{(2500)(4\pi \times 10^{-7})(0.015 \text{ m}^2)}$$

$$= 27{,}600 \text{ A} \cdot \text{turns/Wb}$$

Therefore, the total reluctance in the core is

$$\begin{aligned}\mathcal{R}_{eq} &= \mathcal{R}_1 + \mathcal{R}_2 \\ &= 14{,}300 \text{ A} \cdot \text{turns/Wb} + 27{,}600 \text{ A} \cdot \text{turns/Wb} \\ &= 41{,}900 \text{ A} \cdot \text{turns/Wb}\end{aligned}$$

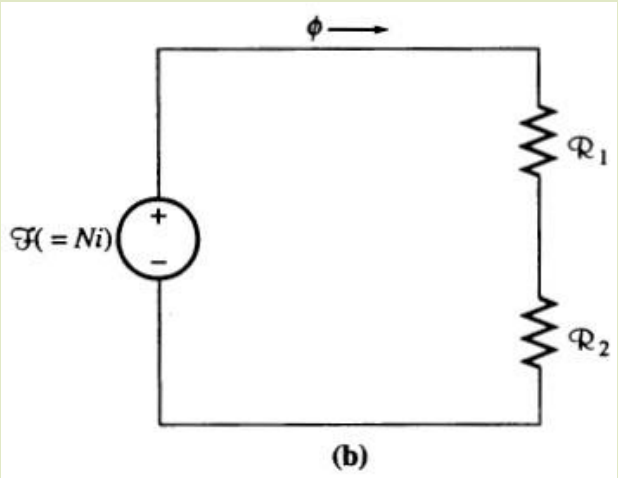
The total magnetomotive force is

$$\mathcal{F} = Ni = (200 \text{ turns})(1.0 \text{ A}) = 200 \text{ A} \cdot \text{turns}$$

The total flux in the core is given by

$$\phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{200 \text{ A} \cdot \text{turns}}{41{,}900 \text{ A} \cdot \text{turns/Wb}}$$

$$= 0.0048 \text{ Wb}$$



Example 1-1

MATLAB script

```
% M-file: ex1_1.m
% M-file to calculate the flux in Example 1-1.
l1 = 0.45;                          % Length of region 1
l2 = 1.3;                           % Length of region 2
a1 = 0.01;                          % Area of region 1
a2 = 0.015;                         % Area of region 2
ur = 2500;                          % Relative permeability
u0 = 4*pi*1E-7;                     % Permeability of free space
n = 200;                            % Number of turns on core
i = 1;                              % Current in amps

% Calculate the first reluctance
r1 = l1 / (ur * u0 * a1);
disp (['r1 = ' num2str(r1)]);

% Calculate the second reluctance
r2 = l2 / (ur * u0 * a2);
disp (['r2 = ' num2str(r2)]);

% Calculate the total reluctance
rtot = r1 + r2;

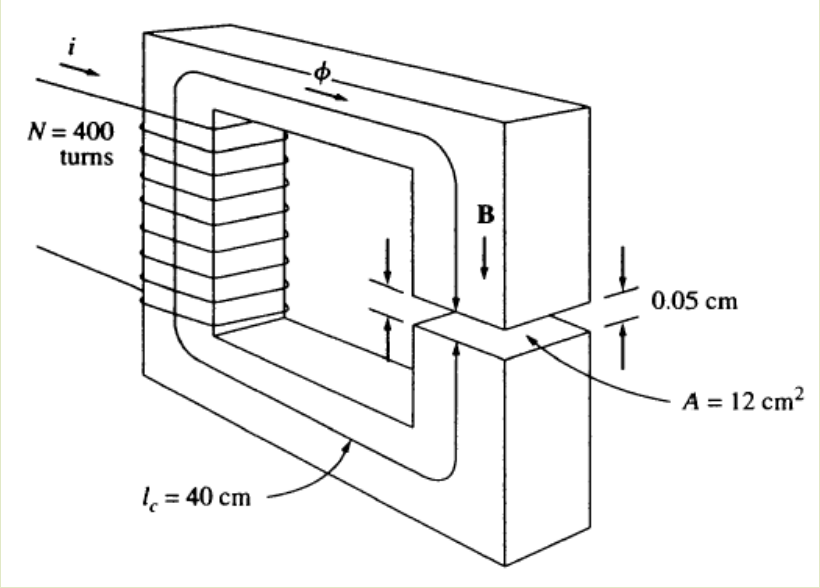
% Calculate the mmf
mmf = n * i;

% Finally, get the flux in the core
flux = mmf / rtot;

% Display result
disp (['Flux = ' num2str(flux)]);
```

Example 1-2

Example 1–2. Figure 1–8a shows a ferromagnetic core whose mean path length is 40 cm. There is a small gap of 0.05 cm in the structure of the otherwise whole core. The cross-sectional area of the core is 12 cm^2, the relative permeability of the core is 4000, and the coil of wire on the core has 400 turns. Assume that fringing in the air gap increases the effective cross-sectional area of the air gap by 5 percent. Given this information, find (a) the total reluctance of the flux path (iron plus air gap) and (b) the current required to produce a flux density of 0.5 T in the air gap.



Solution

Solution

The magnetic circuit corresponding to this core is shown in Figure 1–8b.

(a) The reluctance of the core is

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} = \frac{l_c}{\mu_r \mu_0 A_c} \qquad (1\text{–}32)$$

$$= \frac{0.4 \text{ m}}{(4000)(4\pi \times 10^{-7})(0.002 \text{ m}^2)}$$

$A_c = 0.0012 \text{ m}^2$

$$= 66{,}300 \text{ A} \cdot \text{turns/Wb}$$

$R_c = 66{,}315$ A.turns/Wb

The effective area of the air gap is $1.05 \times 12 \text{ cm}^2 = 12.6 \text{ cm}^2$, so the reluctance of the air gap is

$$\mathcal{R}_a = \frac{l_a}{\mu_0 A_a} \qquad (1\text{–}32)$$

$$= \frac{0.0005 \text{ m}}{(4\pi \times 10^{-7})(0.00126 \text{ m}^2)}$$

$$= 316{,}000 \text{ A} \cdot \text{turns/Wb}$$

Solution

(a) The ferromagnetic core of Example 1–2. (b) The magnetic circuit corresponding to (a).

Therefore, the total reluctance of the flux path is

$$\begin{aligned}\mathcal{R}_{eq} &= \mathcal{R}_c + \mathcal{R}_a \\ &= 66{,}300 \text{ A} \cdot \text{turns/Wb} + 316{,}000 \text{ A} \cdot \text{turns/Wb} \\ &= 382{,}300 \text{ A} \cdot \text{turns/Wb}\end{aligned}$$

R_{eq}=382,315A.turns/Wb

Note that the air gap contributes most of the reluctance even though it is 800 times shorter than the core.

(b) Equation (1–28) states that

$$\mathcal{F} = \phi\mathcal{R} \qquad (1\text{–}28)$$

Since the flux $\phi = BA$ and $\mathcal{F} = Ni$, this equation becomes

$$Ni = BA\mathcal{R}$$

Solution

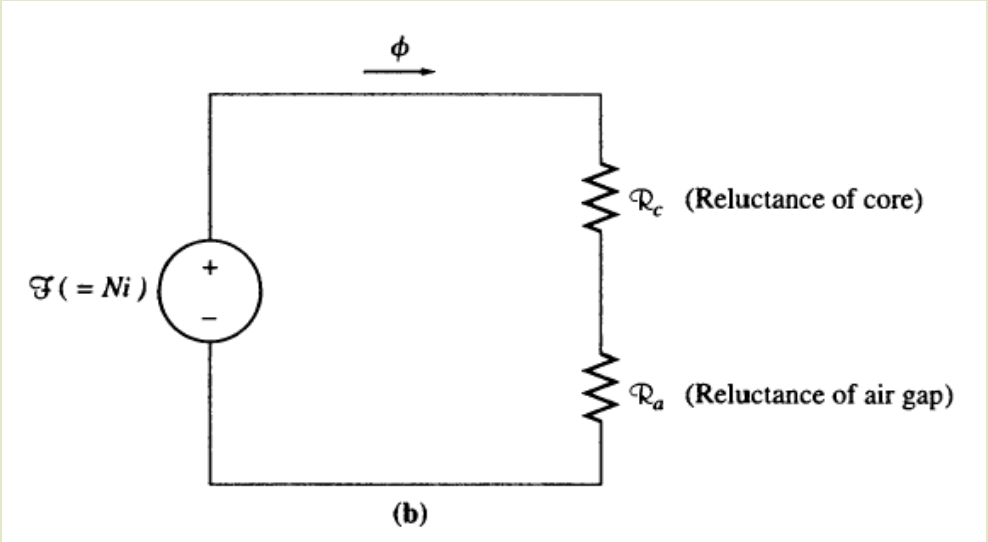
so

$$i = \frac{BA\mathcal{R}}{N}$$

$$= \frac{(0.5\ \text{T})(0.00126\ \text{m}^2)(383{,}200\ \text{A} \cdot \text{turns/ Wb})}{400\ \text{turns}}$$

$$= 0.602\ \text{A}$$ i=0.6037A

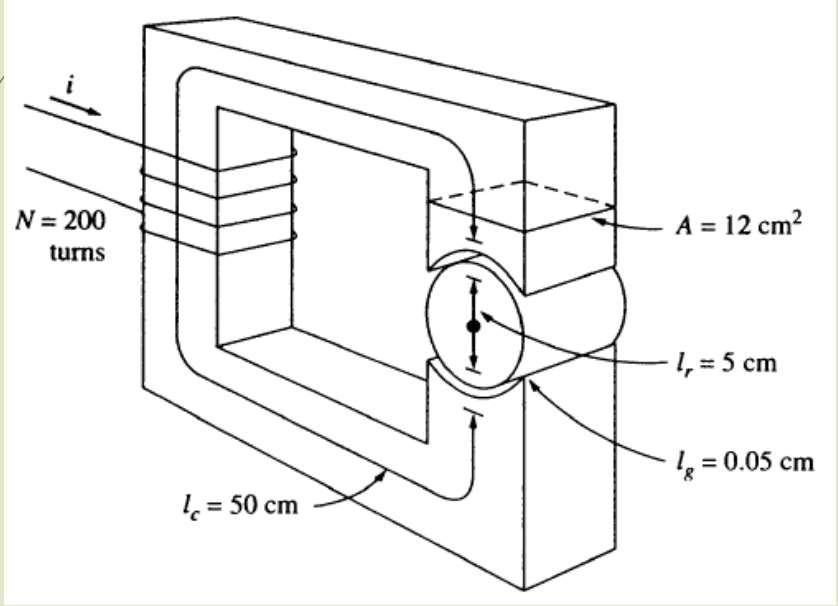
Notice that, since the *air-gap* flux was required, the effective air-gap area was used in the above equation.



Example 1-3

Example 1–3. Figure 1–9a shows a simplified rotor and stator for a dc motor. The mean path length of the stator is 50 cm, and its cross-sectional area is 12 cm^2. The mean path length of the rotor is 5 cm, and its cross-sectional area also may be assumed to be 12 cm^2. Each air gap between the rotor and the stator is 0.05 cm wide, and the cross-sectional area of each air gap (including fringing) is 14 cm^2. The iron of the core has a relative permeability of 2000, and there are 200 turns of wire on the core. If the current in the wire is adjusted to be 1 A, what will the resulting flux density in the air gaps be?

B_g ?



Solution 1-13

Solution

To determine the flux density in the air gap, it is necessary to first calculate the magnetomotive force applied to the core and the total reluctance of the flux path. With this information, the total flux in the core can be found. Finally, knowing the cross-sectional area of the air gaps enables the flux density to be calculated.

The reluctance of the stator is

$$\mathcal{R}_s = \frac{l_s}{\mu_r \mu_0 A_s}$$

$$= \frac{0.5 \text{ m}}{(2000)(4\pi \times 10^{-7})(0.0012 \text{ m}^2)}$$

$$= 166{,}000 \text{ A} \cdot \text{turns/Wb}$$

The reluctance of the rotor is

$$\mathcal{R}_r = \frac{l_r}{\mu_r \mu_0 A_r}$$

$$= \frac{0.05 \text{ m}}{(2000)(4\pi \times 10^{-7})(0.0012 \text{ m}^2)}$$

$$= 16{,}600 \text{ A} \cdot \text{turns/Wb}$$

Solution 1-13

The reluctance of the air gaps is

$$\mathcal{R}_a = \frac{l_a}{\mu_r \mu_0 A_a}$$

$$= \frac{0.0005 \text{ m}}{(1)(4\pi \times 10^{-7})(0.0014 \text{ m}^2)}$$

$$= 284{,}000 \text{ A} \cdot \text{turns/Wb}$$

The magnetic circuit corresponding to this machine is shown in Figure 1–9b. The total reluctance of the flux path is thus

$$\mathcal{R}_{eq} = \mathcal{R}_s + \mathcal{R}_{a1} + \mathcal{R}_r + \mathcal{R}_{a2}$$

$$= 166{,}000 + 284{,}000 + 16{,}600 + 284{,}000 \text{ A} \cdot \text{turns/Wb}$$

$$= 751{,}000 \text{ A} \cdot \text{turns/Wb}$$

The net magnetomotive force applied to the core is

$$\mathcal{F} = Ni = (200 \text{ turns})(1.0 \text{ A}) = 200 \text{ A} \cdot \text{turns}$$

Therefore, the total flux in the core is

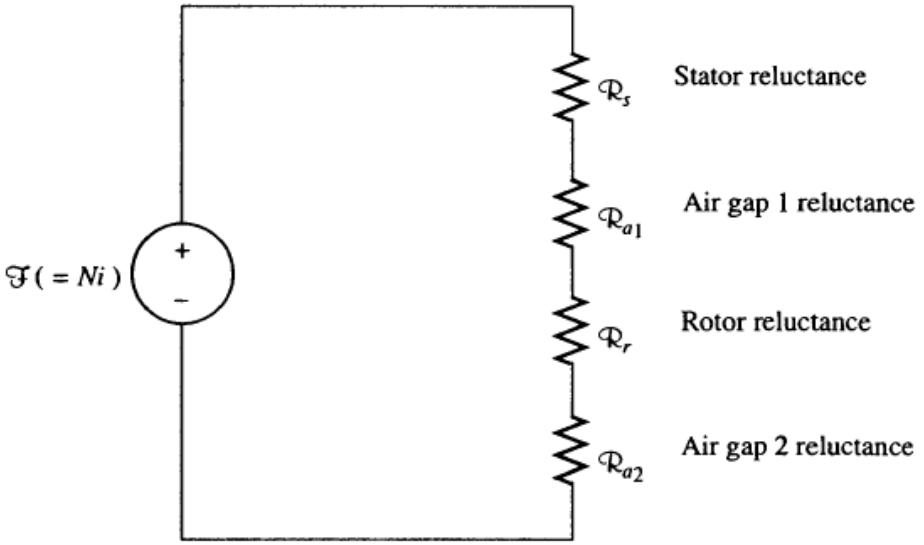
Solution 1-13

$$\phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{200 \text{ A} \cdot \text{turns}}{751{,}000 \text{ A} \cdot \text{turns/Wb}}$$

$$= 0.00266 \text{ Wb}$$

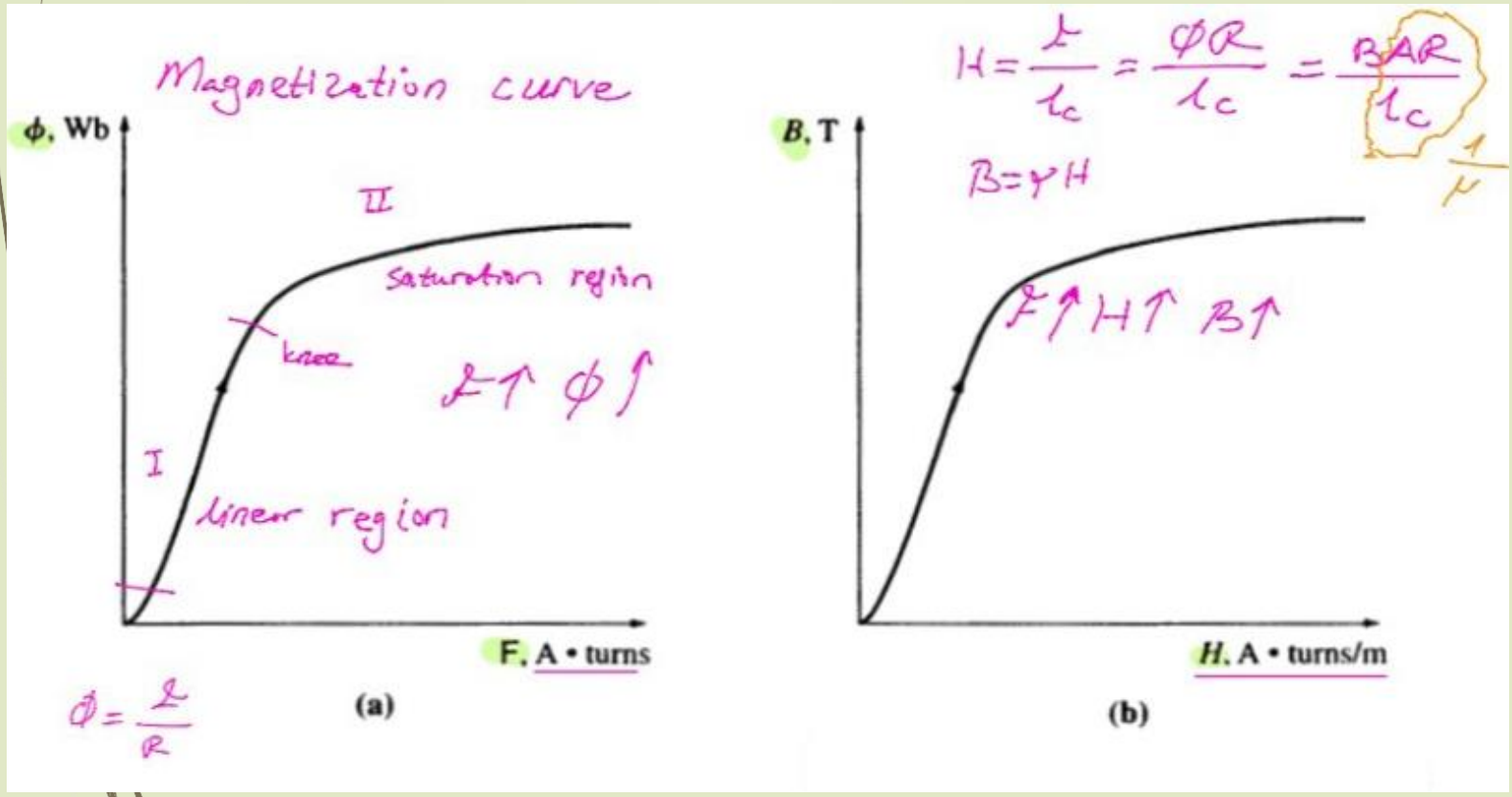
Finally, the magnetic flux density in the motor's air gap is

$$B = \frac{\phi}{A} = \frac{0.000266 \text{ Wb}}{0.0014 \text{ m}^2} = 0.19 \text{ T}$$

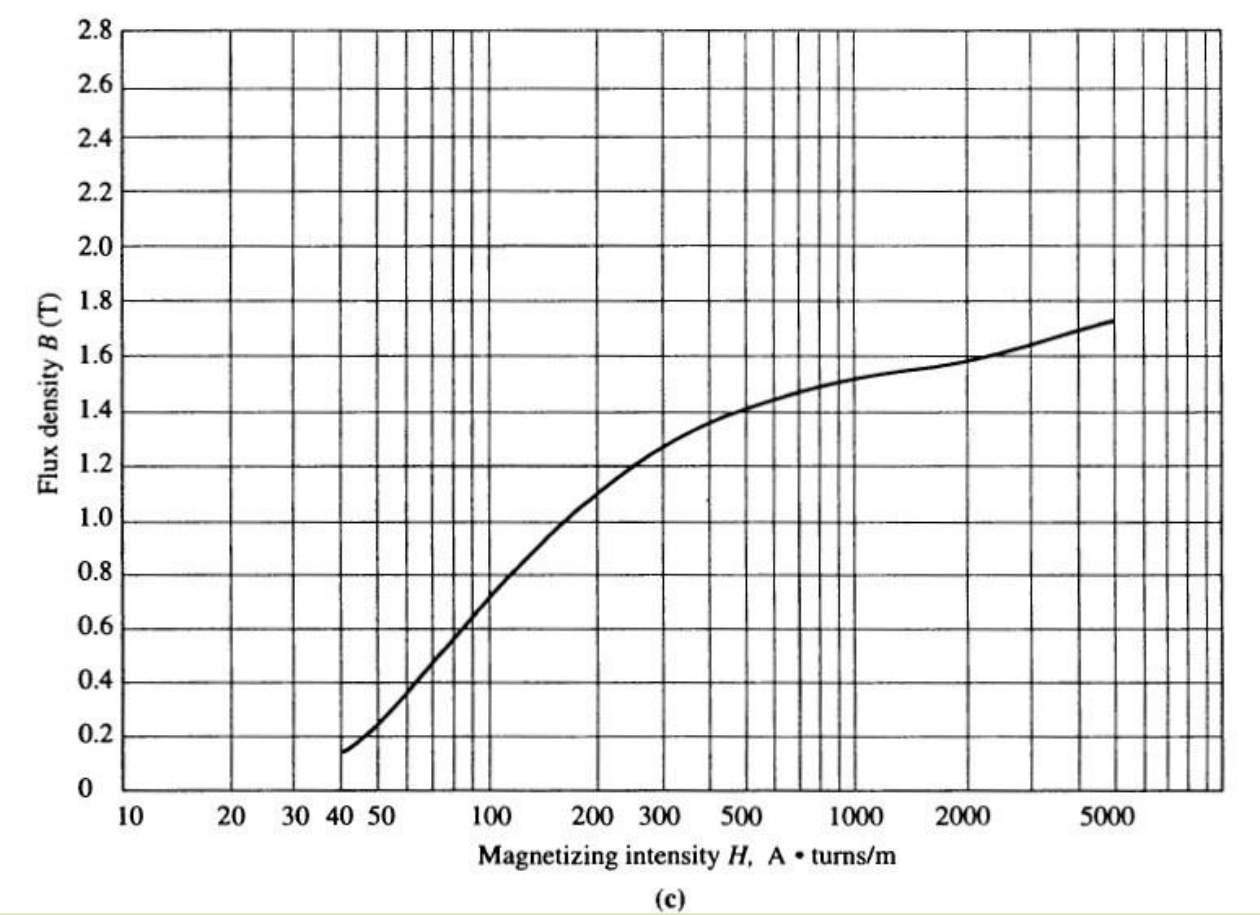


(b)

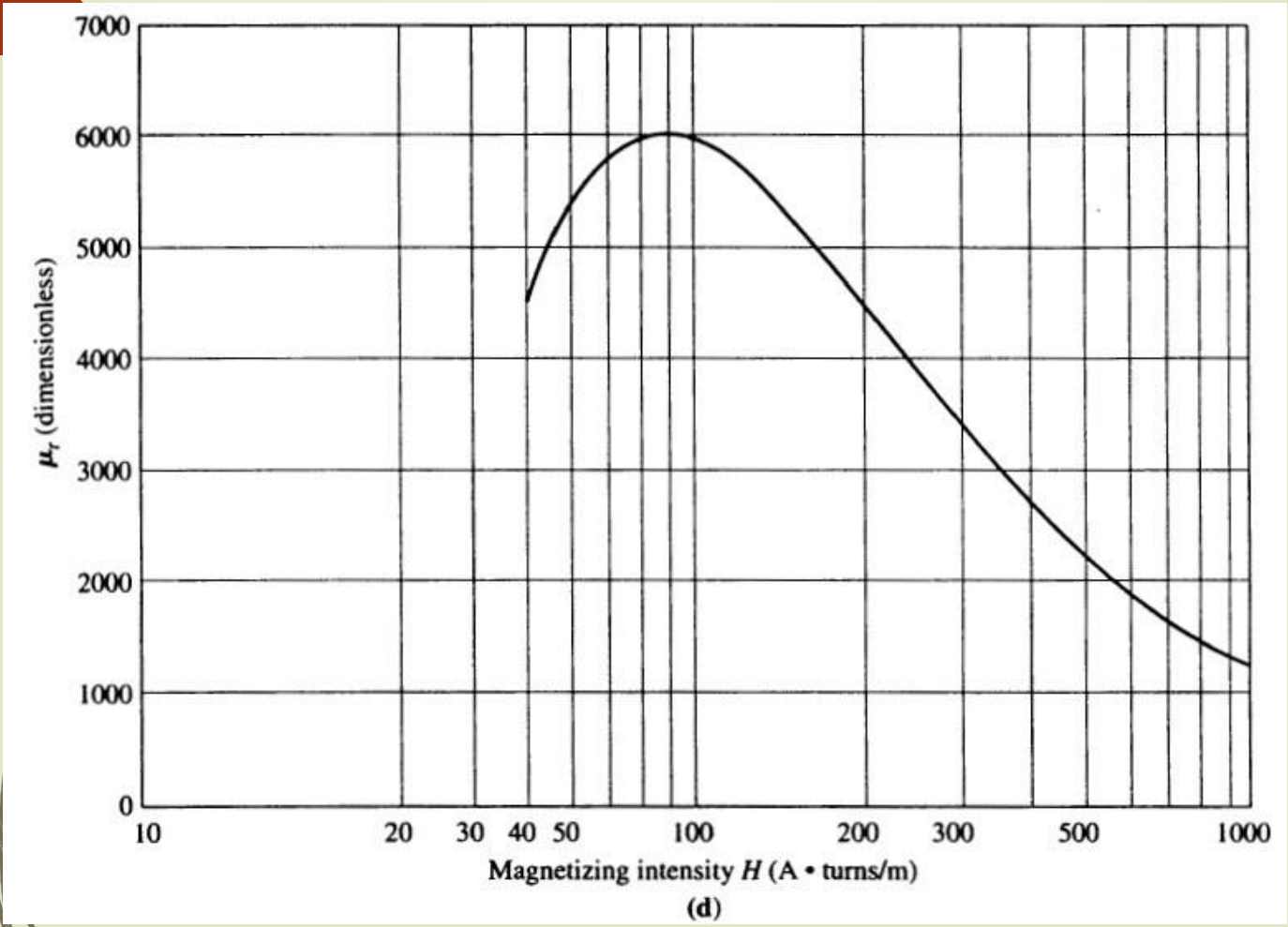
Magnetic behavior of ferromagnetic material - Saturation



Magnetic curve for a typical steel



A plot of relative permeability μ_r



Production of a Magnetic Field

- If we linearize (approximate) the **B-H curve** of the material for a given region or operating range), μ_r **can be thought as constant !**

Material	Relative Permeability
Copper	0.9999906
Silver	0.9999736
Lead	0.9999831
Air	1.00000037
Oxygen	1.000002
Aluminum	1.000021
Titanium 6-4 (Grade 5)	1.00005
Palladium	1.0008
Platinum	1.0003
Manganese	1.001
Cobalt	250
Nickel	600
Iron	280,000

→ **Diamagnetic material (**repelled by a magnetic field**)**

→ **Non-magnetic material (**they are not magnetized**)**

→ **Paramagnetic material (**they become magnetized in a magnetic field but their magnetism disappears when the field is removed**)**

Ferromagnetic materials (attracted by a magnetic field and they can retain their magnetic properties even the magnetic field is removed**)**

Example 1-5

Example 1–5. A square magnetic core has a mean path length of 55 cm and a cross-sectional area of 150 cm^2. A 200-turn coil of wire is wrapped around one leg of the core. The core is made of a material having the magnetization curve shown in Figure 1–10c.

(a) How much current is required to produce 0.012 Wb of flux in the core?
(b) What is the core's relative permeability at that current level?
(c) What is its reluctance?

Solution

(a) The required flux density in the core is

$$B = \frac{\phi}{A} = \frac{0.012 \text{ Wb}}{0.015 \text{ m}^2} = 0.8 \text{ T}$$

From Figure 1–10c, the required magnetizing intensity is

$$H = 115 \text{ A} \cdot \text{turns/m}$$

From Equation (1–20), the magnetomotive force needed to produce this magnetizing intensity is

$$\mathcal{F} = Ni = Hl_c$$
$$= (115 \text{ A} \cdot \text{turns/m})(0.55 \text{ m}) = 63.25 \text{ A} \cdot \text{turns}$$

so the required current is

$$i = \frac{\mathcal{F}}{N} = \frac{63.25 \text{ A} \cdot \text{turns}}{200 \text{ turns}} = 0.316 \text{ A}$$

Solution 1-15

(b) The core's permeability at this current is

$$\mu = \frac{B}{H} = \frac{0.8\ \text{T}}{115\ \text{A} \cdot \text{turns/m}} = 0.00696\ \text{H/m}$$

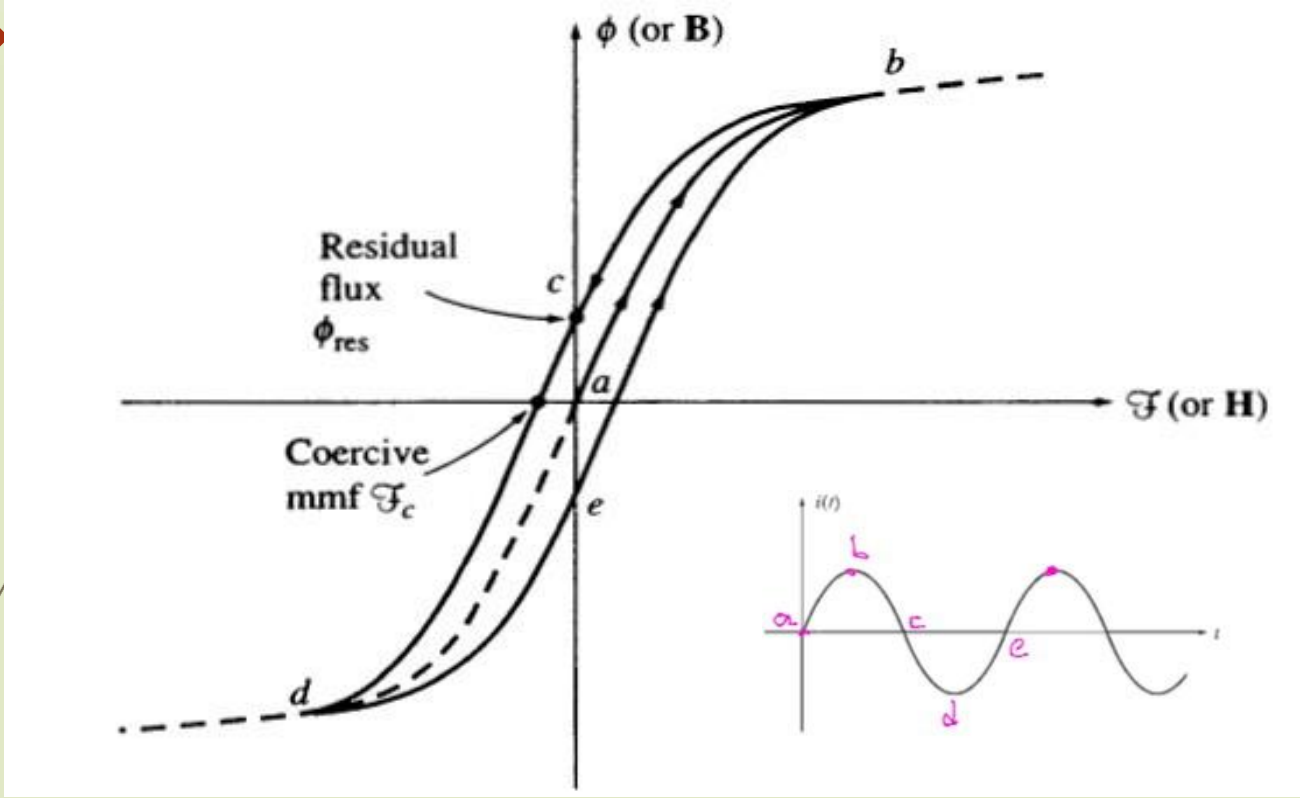
Therefore, the relative permeability is

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.00696\ \text{H/m}}{4\pi \times 10^{-7}\ \text{H/m}} = 5540$$

(c) The reluctance of the core is

$$\mathcal{R} = \frac{\mathcal{F}}{\phi} = \frac{63.25\ \text{A} \cdot \text{turns}}{0.012\ \text{Wb}} = 5270\ \text{A} \cdot \text{turns/Wb}$$

Energy loss in ferromagnetic core – hysteresis loss

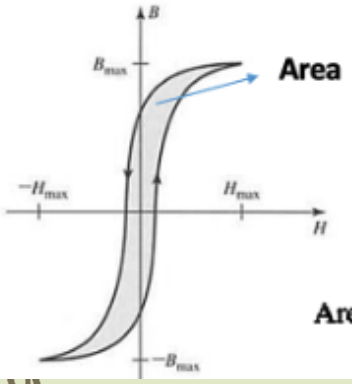


During the cycle of magnetization and demagnetization of magnetic substances, energy is spent and this is spent energy appears in the form of heat. This heat loss is known as hysteresis loss.

Source: https://byjus.com/jee/hysteresis/

Hysteresis loss

- **Turning and reorientation** the domains during each cycle of the alternating current in the ferromagnetic material requires some energy.
- This energy requirement is known as «**hysteresis loss**» in the ferromagnetic materials.
- **Hysteresis loss** causes **heating** in the core (*ferromagnetic*) materials.
- **Hysteresis loss** occurs in all **electrical machines** and **transformers**.
- The area of the **hysteresis loop** is **directly proportional** to the **hysteresis loss**.
- The smaller the applied current or MMF, the smaller the area of the resulting hysteresis loop.



Area ∝ hysteresis loss

Derivation of Hysteresis Loss

- Assume the flux in the core is **sinusoidal**:

$$\emptyset(t) = \emptyset_{max} \sin(wt) = A_c B_{max} \sin(wt)$$

where

$\emptyset_{max}$ is the peak (*max*) value of flux
B_{max} is the peak (*max*) value of magnetic flux density
$w = 2\pi f$ is the angular frequency of the flux

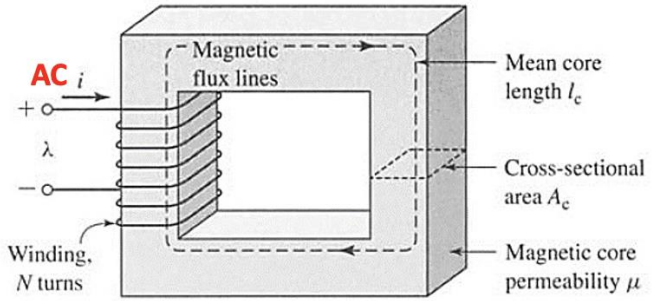


Figure 1.1 Simple magnetic circuit.

- The induced voltage in the winding is; (*Applying Faraday's Law*)

$$e(t) = N\frac{d\phi}{dt}$$

$$e(t) = N\frac{d}{dt}(A_c B_{max} \sin(wt))$$

$$e(t) = \underbrace{NwA_c B_{max}}_{e_{max}} \cos(wt)$$

$$\Rightarrow \quad e_{rms} = \frac{N2\pi f A_c B_{max}}{\sqrt{2}} = \sqrt{2} N\pi f A_c B_{max}$$ (Rms value of induced voltage)

Derivation of Hysteresis Loss

- Remembering the following equation:

$$H = \frac{N.i}{l_c} \Rightarrow i_{rms} = \frac{H_{rms}.\, l_c}{N}$$

- The volt-ampere (*apparent*) power consumed by the core:

$$s = e_{rms}.\, i_{rms}$$

$$s = \sqrt{2}N\pi f A_c B_{max}.\ \frac{H_{rms}.\, l_c}{N}$$

$$s = \sqrt{2}\pi f B_{max}.\, H_{rms}(A_c l_c)$$

$A_c l_c$: core volume

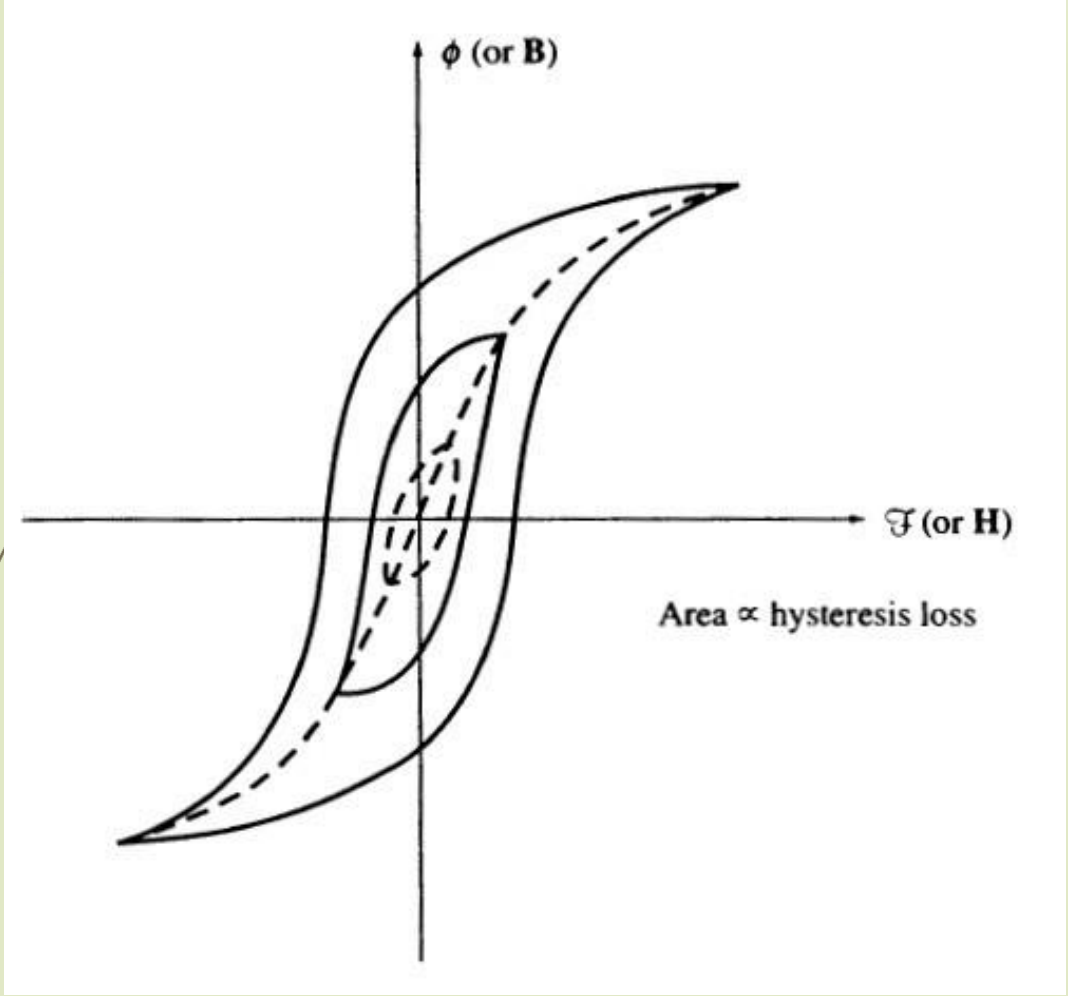
- s is also called as «**rms exciting voltamperes required to excite the core**»
- So **hysteresis loss** is proportional to the
  1) Frequency of excitation
  2) Peak flux density
  3) Rms magnetic field intensity
  4) Core volume
- s can also be calculated in terms of **per unit mass,**

$$s_a = \frac{\sqrt{2}\pi f B_{max}.\, H_{rms}(A_c l_c)}{mass\ of\ the\ core} \Rightarrow \rho_c A_c l_c$$

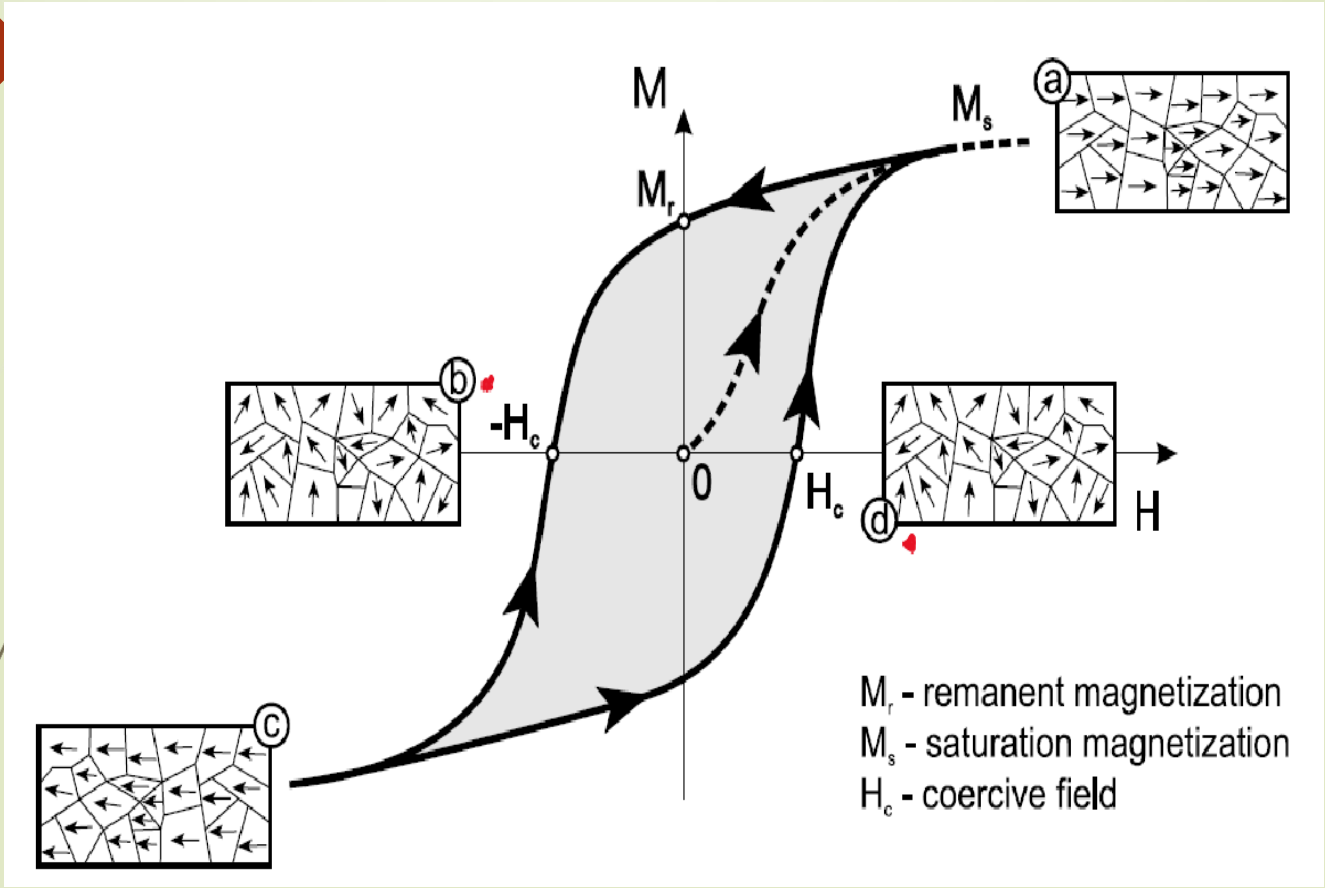
$$s_a = \frac{\sqrt{2}\pi f B_{max}.\, H_{rms}}{\rho_c}$$

ρ_c : Core density

The effect of magnetomotive force on the hysteresis loop



Magnetization curve

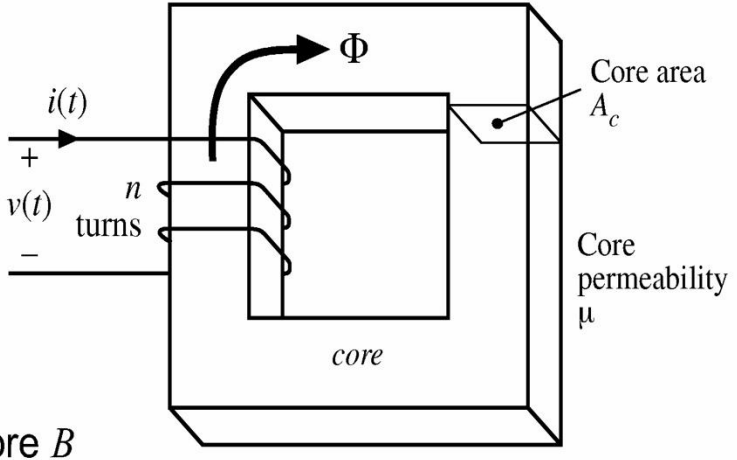


Several applications of hysteresis are found in ferromagnets. It is mostly used to retain a memory, for example, hard disks, magnetic tape, and credit cards.

Hysteresis loss

Energy per cycle W flowing into n-turn winding of an inductor, excited by periodic waveforms of frequency f:

$$W = \int_{\text{one cycle}} v(t)i(t)\,dt$$



Relate winding voltage and current to core B and H via Faraday's law and Ampere's law:

$$v(t) = nA_c \frac{dB(t)}{dt} \qquad H(t)\ell_m = ni(t)$$

Substitute into integral:

$$W = \int_{\text{one cycle}} \left(nA_c \frac{dB(t)}{dt}\right)\left(\frac{H(t)\ell_m}{n}\right)dt$$

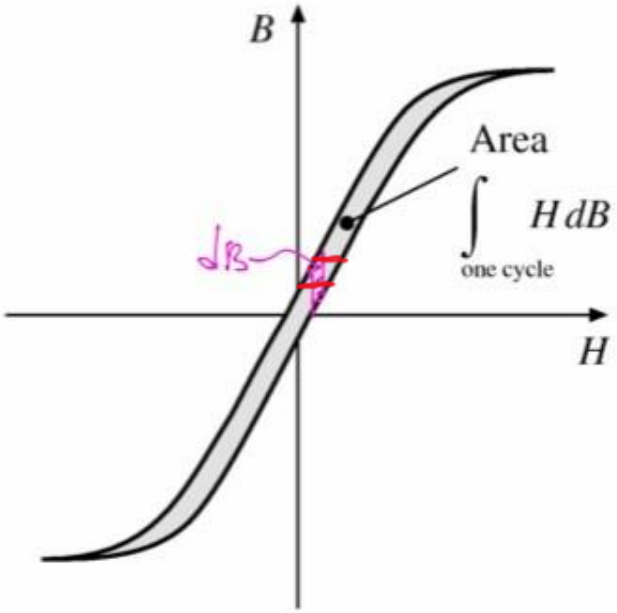
$$= \left(A_c\ell_m\right)\int_{\text{one cycle}} H\,dB$$

Hysteresis loss

$$W = (A_c \ell_m) \int_{\text{one cycle}} H\,dB$$

Volume of the core

The term $A_c \ell_m$ is the volume of the core, while the integral is the area of the B–H loop.



(energy lost per cycle) = (core volume) (area of B–H loop)

$P = \frac{W}{t}$

$\frac{1}{t}$

$$P_H = (f)(A_c \ell_m) \int_{\text{one cycle}} H\,dB$$

Hysteresis loss is directly proportional to applied frequency

Eddy Current Loss

- As observed from the eddy current loss equation, the amount of energy lost is proportional to the square of the thickness and the volume of the material.

Eddy Current Loss (P_e) = $K_e B_m^2 f^2 t^2 V$

Where,

P_e = Eddy curret loss (Watt)

B_m = Maximum Flux density(W_b/m^2)

f = Frequency of supply (f)

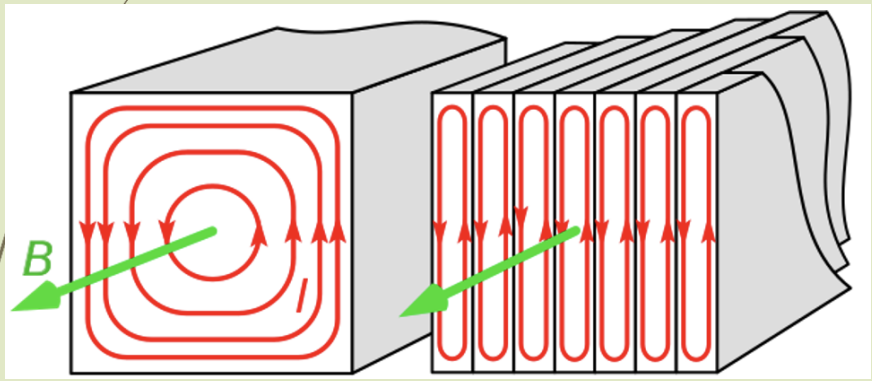
t = Thickness of lamination(meter)

V = Volume of the material (M^3)

K_e = Eddy current Constant

Eddy Current Loss

- For this reason, it is customary to break up the ferromagnetic core into many small laminations.
- An insulating oxide or resin is used between these laminations.
- Because the insulating layers are extremely thin, this action reduces eddy current losses without changing too much the magnetic properties of the core.



Eddy currents in laminated cores (*right*) are smaller than those in solid cores (*left*)

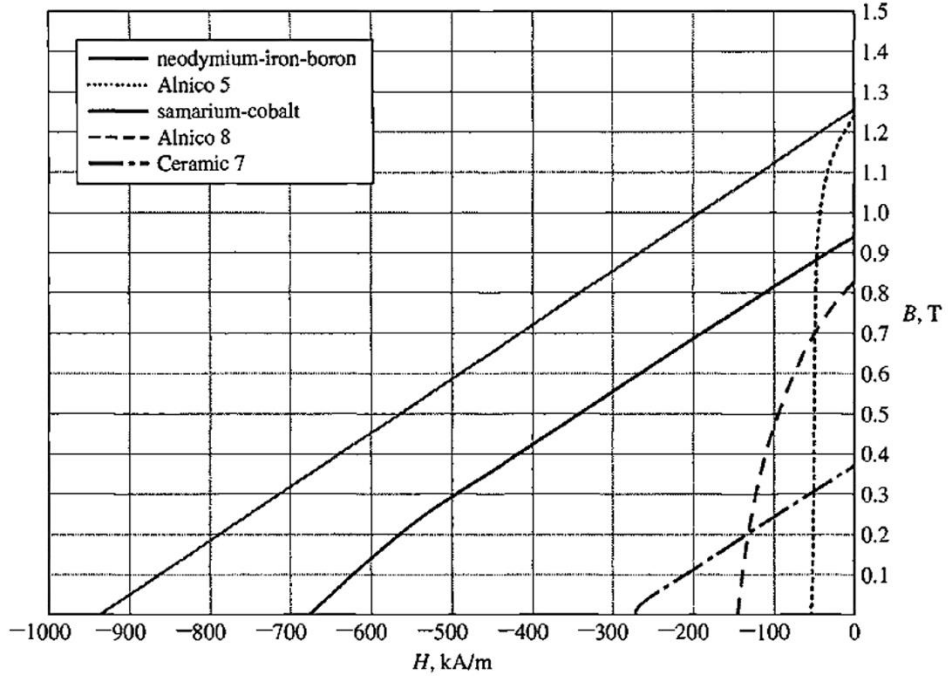
Source: https://www.motioncontroltips.com/hysteresis-loss/

Permanent Magnets

- Permanent magnets have a **large residual flux**.
- Residual flux means that **without excitation** (*no current is applied*), the magnet produces a high flux **by itself**.
- Permanent magnets also have a **large coercive MMF**.
- **Coercive MMF** is the MMF that is required to make the flux in the permanent magnet zero.
- **Coercivity** can be thought of as a measure of the magnitude of the MMF required to **demagnetize** the material.
- Permanent magnets find application in many devices, including loudspeakers, ac and dc motors, microphones, and analog electric meters.

Permanent magnets

Permanent Magnets





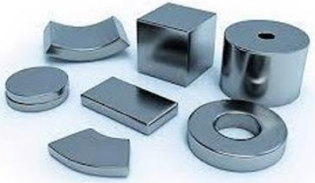
Alnico 5



Samarium-cobalt

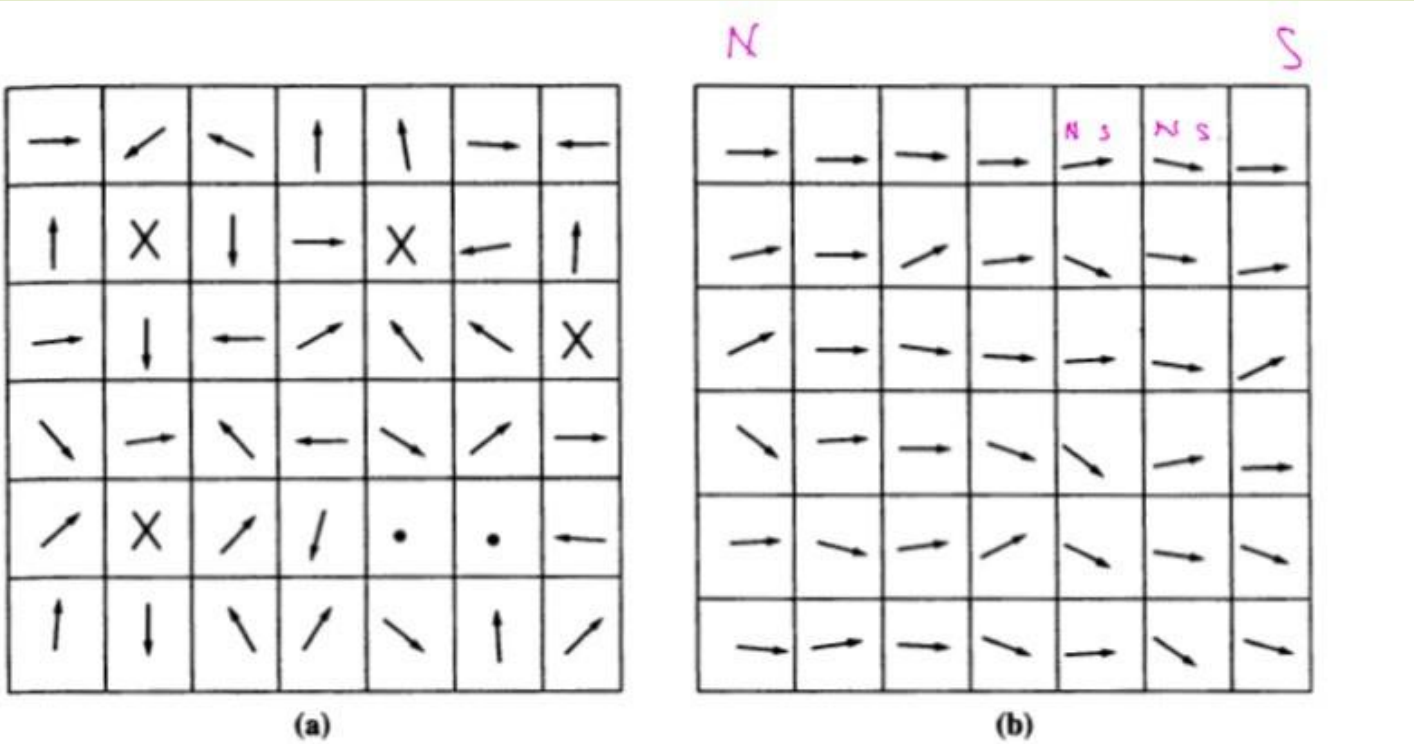


Alnico 8



neodymium iron boron

Hysteresis loop – residual flux



(a) Magnetic domains oriented randomly. (b) Magnetic domains lined up in the presence of an external magnetic field.

Ferromagnetic materials consist of small domains, within which magnetic moments of atoms are aligned. However, magnetic moments of domains are oriented randomly.

Faraday's law: induce voltage from a time-varying magnetic field

1. Induced voltage magnitude and polarity

$$e_{ind} = -\frac{d\phi}{dt}$$

$$e_{ind} = \sum_{i=1}^{N} e_i \qquad \phi = \sum_{i=1}^{N} \phi_i$$

$$e_{ind} = -N\frac{d\phi}{dt} \quad , \quad e_i = \frac{-d\phi_i}{dt}$$

e_{ind} = voltage induced in the coil

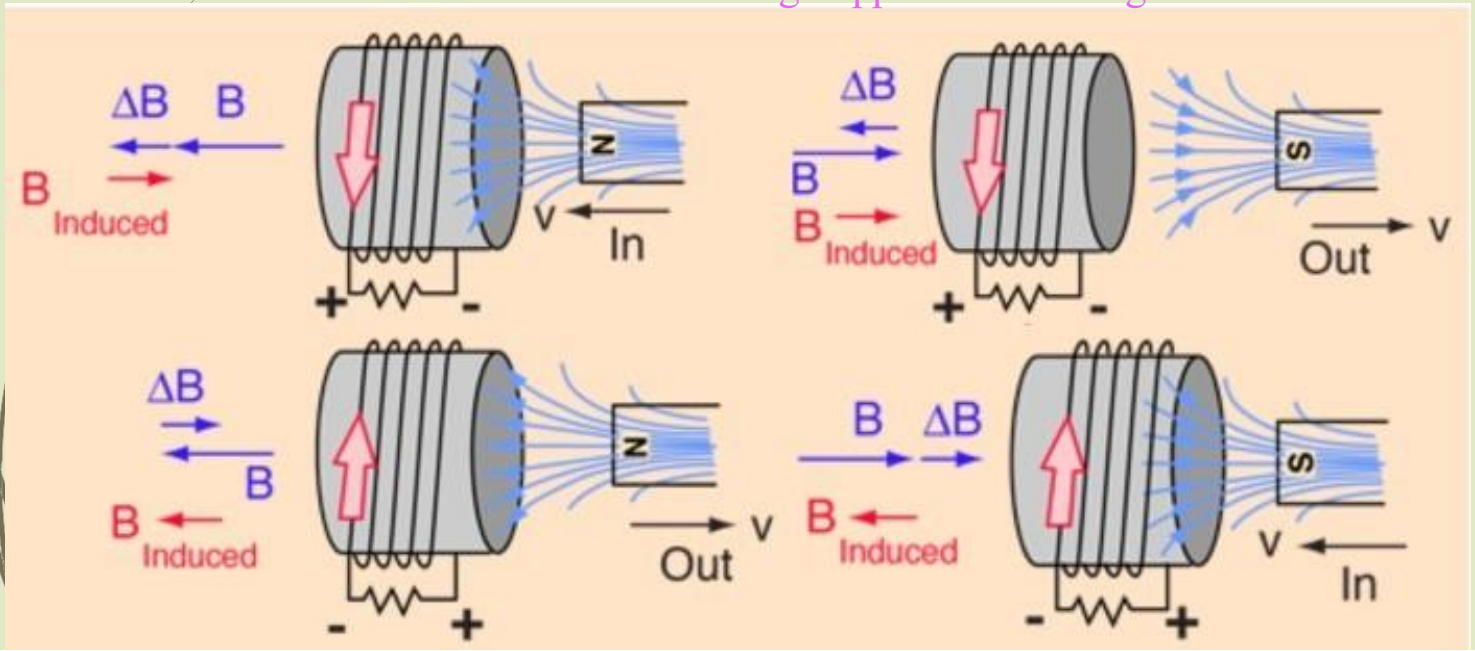
N = number of turns of wire in coil

ϕ = flux passing through coil

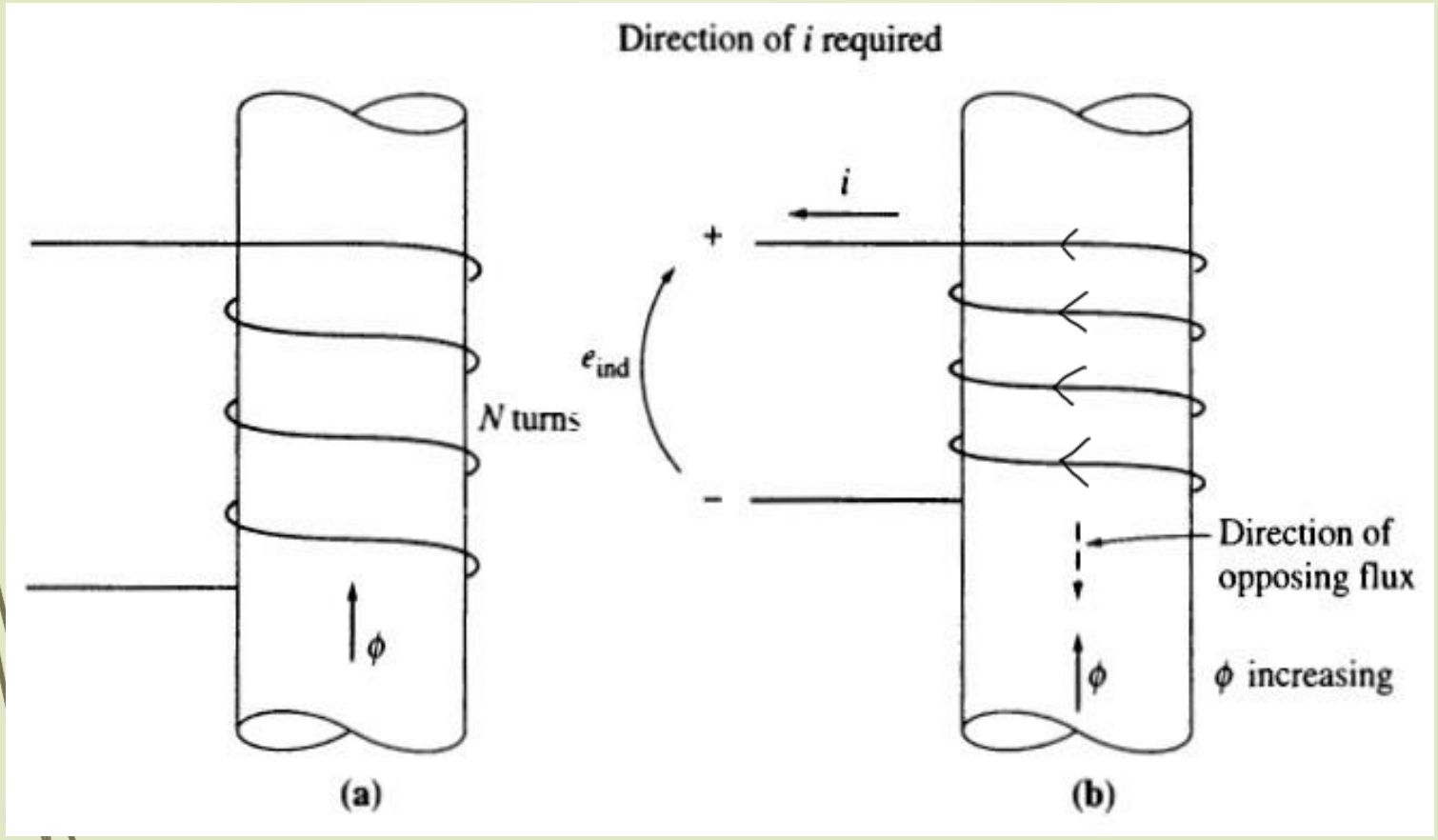
The minus sign in the equations is an expression of *Lenz's law*.

Lenz's law (induced voltage (back emf) and magnetic flux)

When an emf is generated by a change in magnetic flux according to Faraday's Law, the polarity of the induced emf is such that it produces a current whose magnetic field opposes the change which produces it. The induced magnetic field inside any loop of wire always acts to keep the magnetic flux in the loop constant. In the examples below, if the B field is increasing, the induced field acts in opposition to it. If it is decreasing, the induced field acts in the direction of the applied field to try to keep it constant. The direction of this induced voltage opposes the change in current.



The induced voltage polarity – Lenz's law



Flux and flux linkage

The term in parentheses in Equation (1–40) is called *the flux linkage* λ of the coil, and Faraday's law can be rewritten in terms of flux linkage as

$$e_{\text{ind}} = \frac{d\lambda}{dt} \tag{1–41}$$

where

$$\lambda = \sum_{i=1}^{N} \phi_i \tag{1–42}$$

The units of flux linkage are weber-turns.

Example 1-6

Example 1–6. Figure 1–15 shows a coil of wire wrapped around an iron core. If the flux in the core is given by the equation

$$\phi = 0.05 \sin 377t \quad \text{Wb}$$

If there are 100 turns on the core, what voltage is produced at the terminals of the coil? Of what polarity is the voltage during the time when flux is *increasing* in the reference direction shown in the figure? Assume that all the magnetic flux stays within the core (i.e., assume that the flux leakage is zero).

The core of Example 1–6. Determination of the voltage polarity at the terminals is shown.

Example 1-6 (continued)

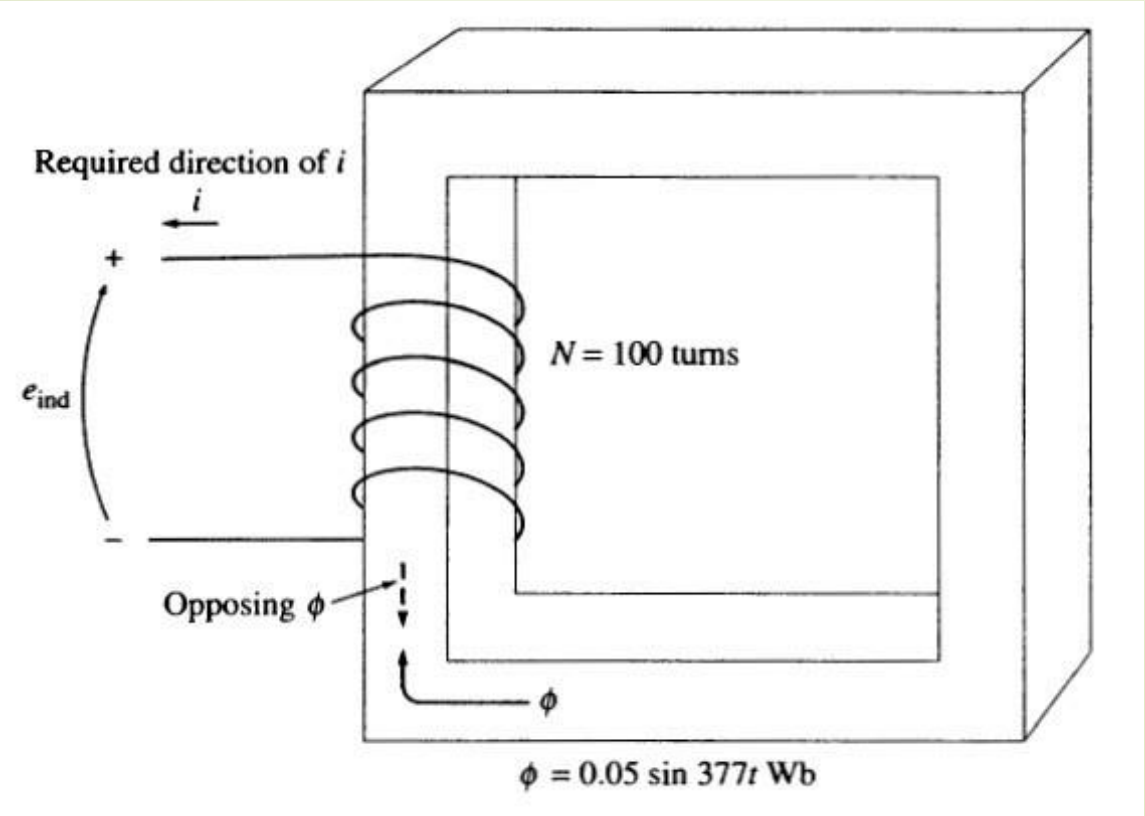


Fig. 1-15

Example 1-6 (continued)

Solution

By the same reasoning as in the discussion on pages 29–30, the direction of the voltage while the flux is increasing in the reference direction must be positive to negative, as shown in Figure 1–15. The *magnitude* of the voltage is given by

$$e_{\text{ind}} = N\frac{d\phi}{dt}$$

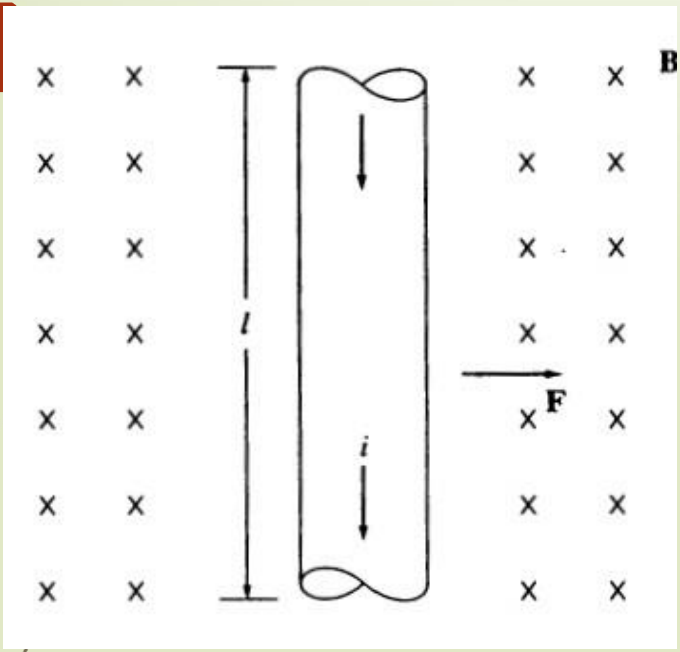
$$= (100 \text{ turns})\frac{d}{dt}(0.05 \sin 377t)$$

$$= 1885 \cos 377t$$

or alternatively,

$$e_{\text{ind}} = 1885 \sin(377t + 90°) \text{ V}$$

Producing induced force on a wire



Using left hand rule

forefinger → B
middle finger → i
thumb → F

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

$$F = ilB \sin \theta$$

i = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

The induced force on a wire by a current in the presence of a magnetic field is the basis of *motor action.*

Example 1-7

Example 1–7. Figure 1–16 shows a wire carrying a current in the presence of a magnetic field. The magnetic flux density is 0.25 T, directed into the page. If the wire is 1.0 m long and carries 0.5 A of current in the direction from the top of the page to the bottom of the page, what are the magnitude and direction of the force induced on the wire?

Solution

The direction of the force is given by the right-hand rule as being to the right. The magnitude is given by

$$F = ilB \sin \theta \qquad (1\text{–}44)$$

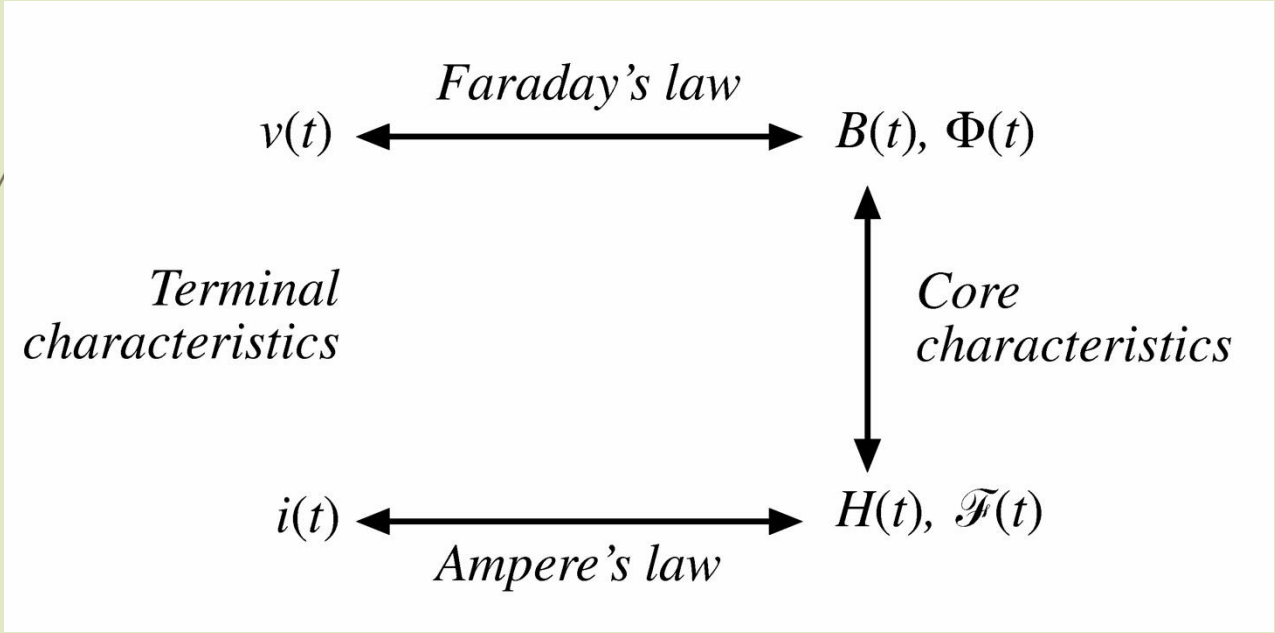
$$= (0.5 \text{ A})(1.0 \text{ m})(0.25 \text{ T}) \sin 90° = 0.125 \text{ N}$$

Therefore,

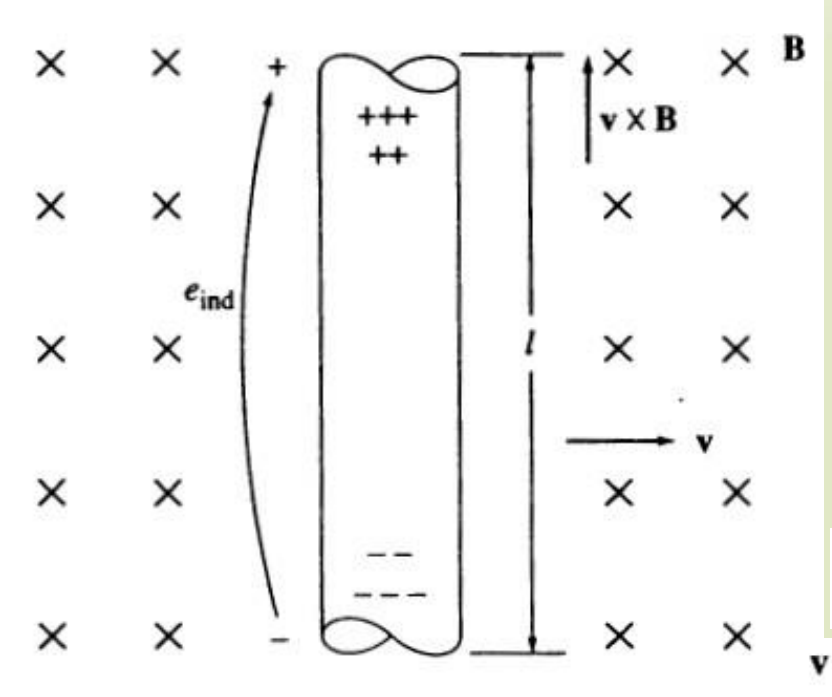
$$\mathbf{F} = 0.125 \text{ N, directed to the right}$$

Relationship between electric-magnetic variables

- Magnetic field: Ampere's law
- Magnetic flux: magnetic material, hysteresis characteristics
- Transformer: Faraday's law, Len's law



Induced voltage on a conductor moving in a magnetic field



Using right-hand rule

positive polarity is the direction of $\vec{v} \times \vec{B}$



$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

$\mathbf{v}$ = velocity of the wire

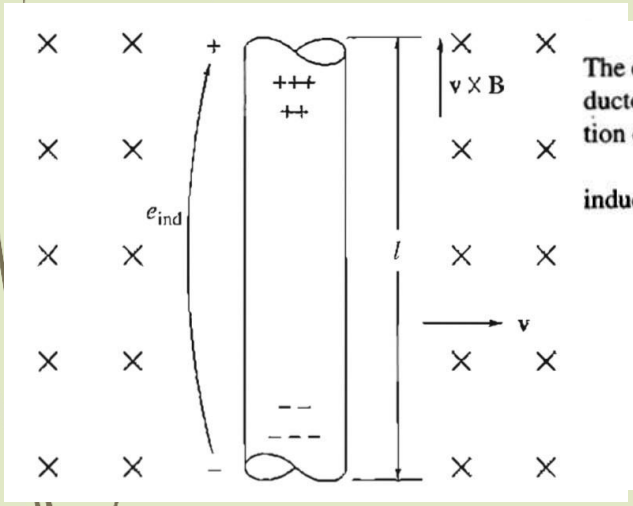
$\mathbf{B}$ = magnetic flux density vector

$\mathbf{l}$ = length of conductor in the magnetic field

The polarity of the induced voltage is determined by the right-hand rule.

Example 1-8

Example 1–8. Figure 1–17 shows a conductor moving with a velocity of 5.0 m/s to the right in the presence of a magnetic field. The flux density is 0.5 T into the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?



Solution

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ in this example is up. Therefore, the voltage on the conductor will be built up positive at the top with respect to the bottom of the wire. The direction of vector **l** is up, so that it makes the smallest angle with respect to the vector $\mathbf{v} \times \mathbf{B}$.

Since **v** is perpendicular to **B** and since $\mathbf{v} \times \mathbf{B}$ is parallel to **l**, the magnitude of the induced voltage reduces to

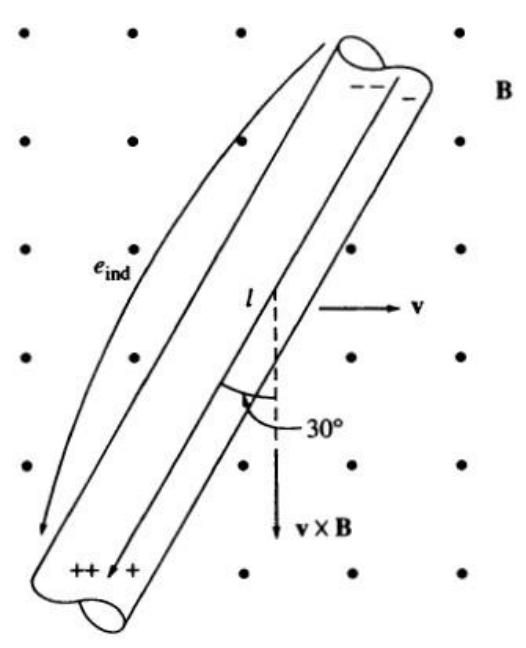
$$\begin{aligned} e_{ind} &= (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45) \\ &= (vB \sin 90°)\, l \cos 0° \\ &= vBl \\ &= (5.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \\ &= 2.5 \text{ V} \end{aligned}$$

Thus the induced voltage is 2.5 V, positive at the top of the wire.

Fig 1.17 A conductor in the presence of a magnetic field

Example 1-9

Example 1–9. Figure 1–18 shows a conductor moving with a velocity of 10 m/s to the right in a magnetic field. The flux density is 0.5 T, out of the page, and the wire is 1.0 m in length, oriented as shown. What are the magnitude and polarity of the resulting induced voltage?



Solution

The direction of the quantity $\mathbf{v} \times \mathbf{B}$ is down. The wire is not oriented on an up-down line, so choose the direction of l as shown to make the smallest possible angle with the direction of $\mathbf{v} \times \mathbf{B}$. The voltage is positive at the bottom of the wire with respect to the top of the wire. The magnitude of the voltage is

$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45)$$

$$= (vB \sin 90°)\, l \cos 30°$$

$$= (10.0 \text{ m/s})(0.5 \text{ T})(1.0 \text{ m}) \cos 30°$$

$$= 4.33 \text{ V}$$

Fig 1.18 A conductor in the presence of a magnetic field

The linear DC machine - a simple example

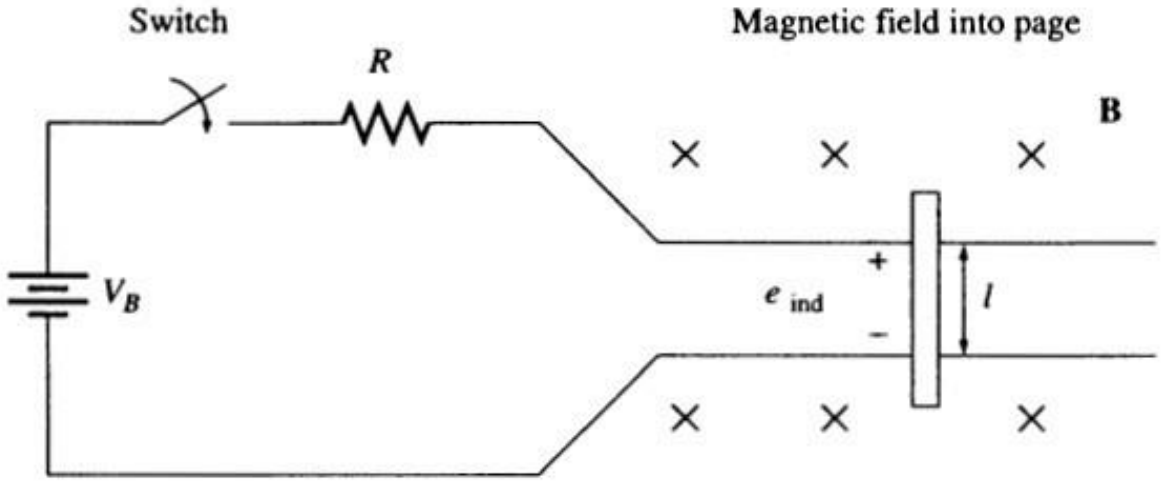


FIGURE 1–19
A linear dc machine. The magnetic field points into the page.

Assume

- Uniform and constant magnetic field
- Frictionless tracks
- A bar lying across the tracks
- Battery
- Switch

The linear DC machine

- The simplest and easiest to understand version of a DC machine.
- It works on the same principles and exhibits the same behavior as real generators and motors.
- Therefore, it serves as a good starting point in the study of machines.

Determining the behavior of the machine with four basic equations

1. The equation for the force on a wire in the presence of a magnetic field:

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B}) \qquad (1\text{–}43)$$

where $\mathbf{F}$ = force on wire

i = magnitude of current in wire

$\mathbf{l}$ = length of wire, with direction of $\mathbf{l}$ defined to be in the direction of current flow

$\mathbf{B}$ = magnetic flux density vector

The linear DC machine

2. The equation for the voltage induced on a wire moving in a magnetic field:

$$\boxed{e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}} \qquad (1\text{–}45)$$

where e_{ind} = voltage induced in wire
$\mathbf{v}$ = velocity of the wire
$\mathbf{B}$ = magnetic flux density vector
$\mathbf{l}$ = length of conductor in the magnetic field

3. Kirchhoff's voltage law for this machine. From Figure 1–19 this law gives

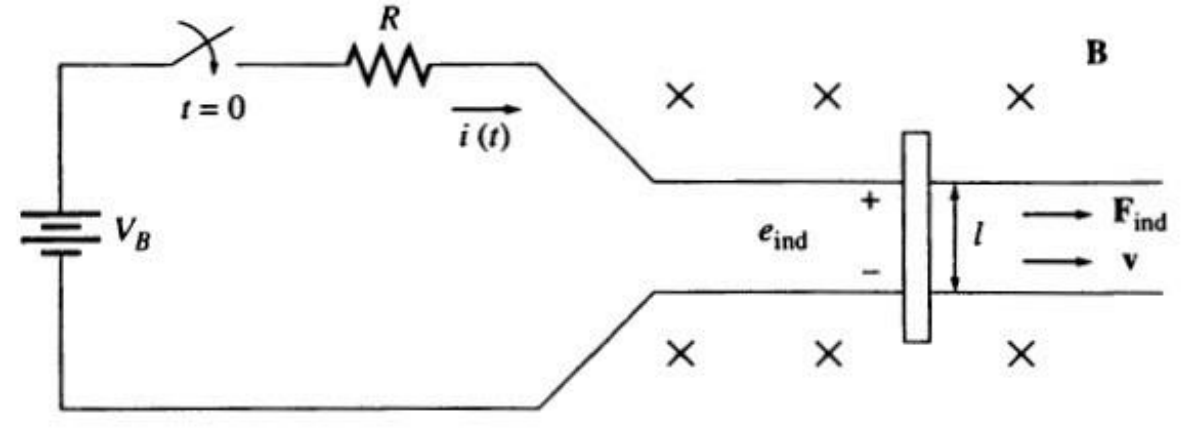
$$V_B - iR - e_{\text{ind}} = 0$$

$$\boxed{V_B = e_{\text{ind}} + iR = 0} \qquad (1\text{–}46)$$

4. Newton's law for the bar across the tracks:

$$\boxed{F_{\text{net}} = ma} \qquad (1\text{–}7)$$

Starting a linear DC machine



1. **Current ⇒** $i = \dfrac{V_B - e_{ind}}{R}$

2. **Induced force ⇒** $F_{ind} = ilB$ to the right

3. **Induced voltage ⇒** $e_{ind} = vBl$ positive upward

$$i\downarrow = \frac{V_B - e_{ind}\uparrow}{R}$$

Starting a linear DC machine

- Eventually the bar will reach a constant steady-state speed where the net force on the bar is zero.

- This occurs when e_{ind} rises until it equals the voltage V_B.
- At that time, the bar will be moving at a speed given by

$$V_B = e_{\text{ind}} = v_{ss} Bl$$

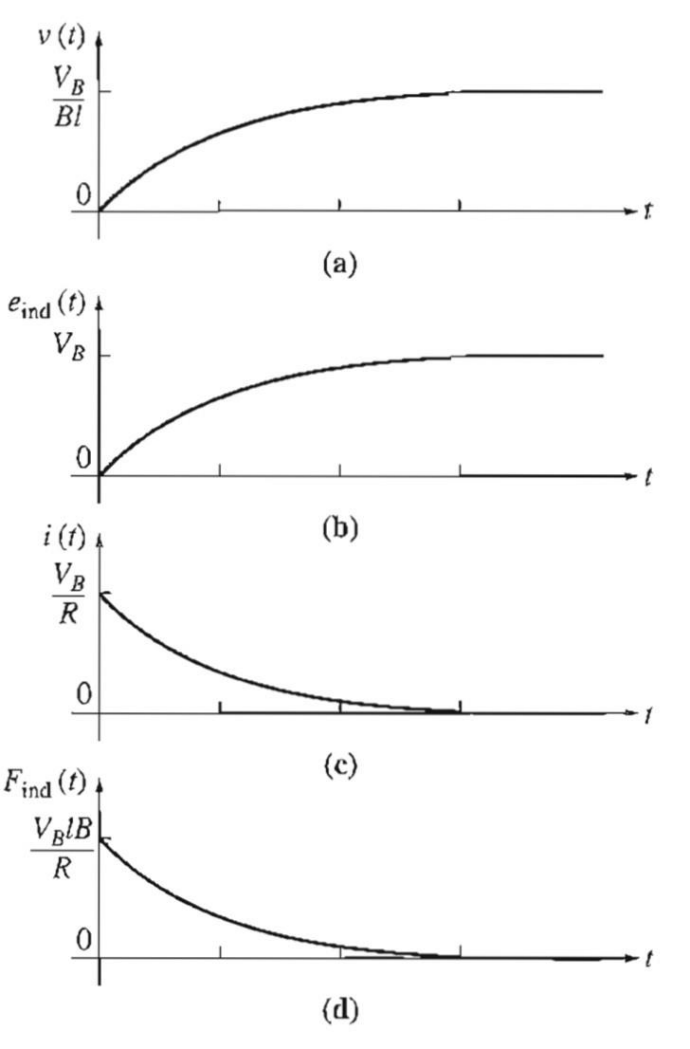
$$v_{ss} = \frac{V_B}{Bl}$$

- The bar will continue to slide along at this no-load speed forever unless some external force disturbs it.

Summarize of starting a dc machine

1. Closing the switch produces a current flow $i = V_B / R$.
2. The current flow produces a force on the bar given by $F = ilB$.
3. The bar accelerates to the right, producing an induced voltage e_{ind} as it speeds up.
4. This induced voltage reduces the current flow $i = (V_B - e_{ind}\uparrow)/R$.
5. The induced force is thus decreased ($F = i\downarrow lB$) until eventually $F = 0$. At that point, $e_{ind} = V_B$, $i = 0$, and the bar moves at a constant no-load speed $v_{ss} = V_B / Bl$.

DC linear machine operates at no-load condition



Velocity v(t) vs time

Velocity e_{ind} (t) vs time

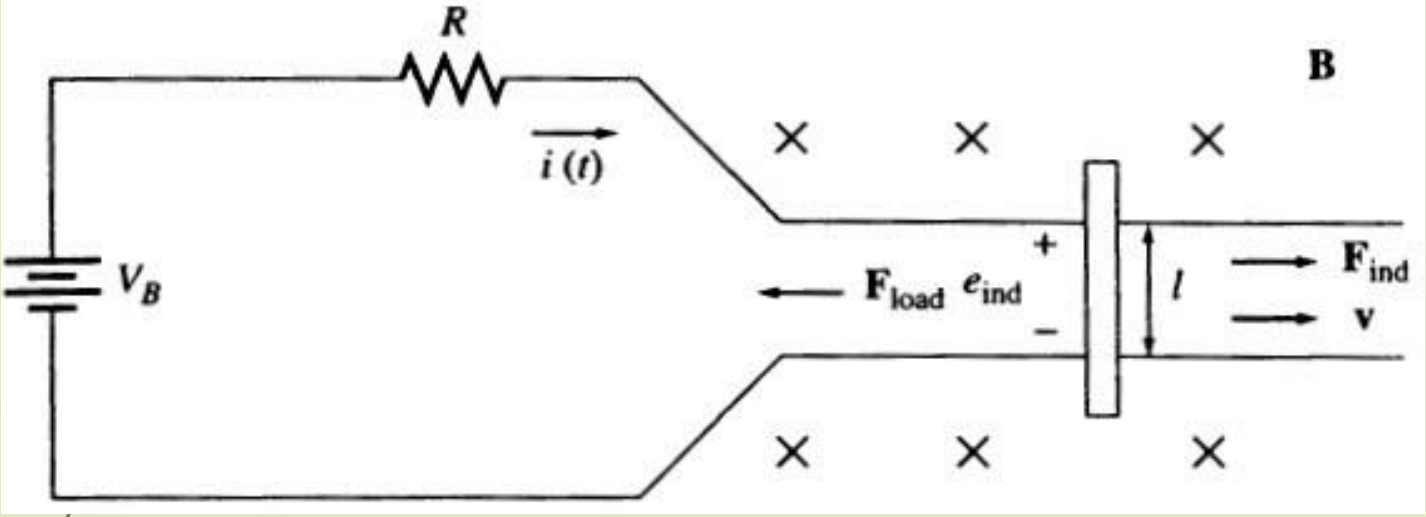
Current i(t) vs time

Induced force F_{ind}(t) vs time

Fig. 1-21

Linear dc machine as a motor

- Applying the load (F_{load})



$$F_{net} = F_{load} - F_{ind}$$

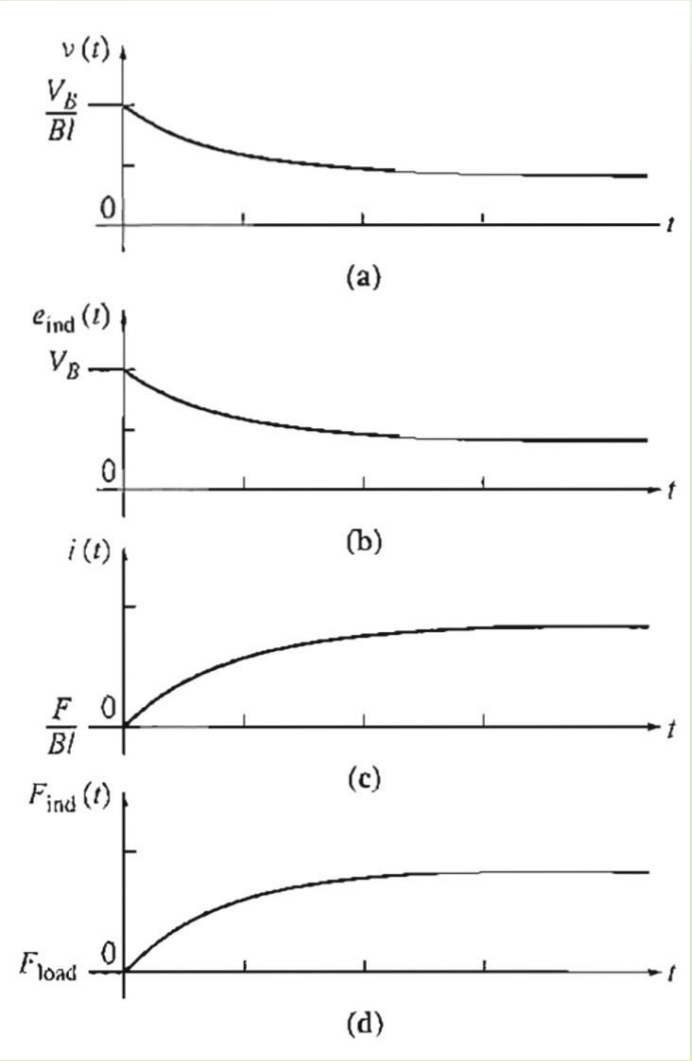
- The conversion of power between mechanical and electrical

$$P_{conv} = e_{ind} i = F_{ind} v$$

Summarize of a dc motor operation

1. A force $\mathbf{F}_{load}$ is applied opposite to the direction of motion, which causes a net force $\mathbf{F}_{net}$ opposite to the direction of motion.
2. The resulting acceleration $a = F_{net}/m$ is negative, so the bar slows down ($v\downarrow$).
3. The voltage $e_{ind} = v\downarrow Bl$ falls, and so $i = (V_B - e_{ind}\downarrow)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a lower speed v.
5. An amount of electric power equal to $e_{ind}i$ is now being converted to mechanical power equal to $F_{ind}v$, and the machine is acting as a motor.

Graphical plots for motor action



Speed v(t) vs time

Induced voltage e_{ind} (t) vs tim

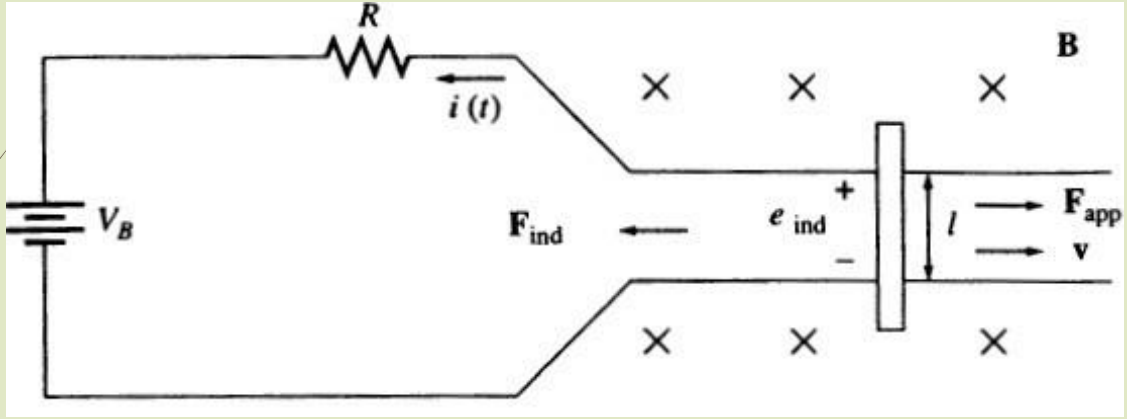
Current i(t) vs time

Induced force F_{ind}(t) vs time

Fig. 1-23

Linear dc machine as a generator

- Applying an external force (F_{app}) on the moving direction



- The conversion of mechanical power into electrical

$$P_{conv} = Fv = ie_{ind}$$

Fnet & Fapp in the direction of motion

Linear dc machine as a generator

- With applying a force to the bar in the direction of motion, the bar positively accelerates and the velocity v of the bar increases.
- As the velocity increases, $e_{ind} = v\uparrow$ will increase and will be larger than the battery voltage V_B.
- When $e_{ind} > V_B$, the current reverses direction and is now given by the equation

$$i = \frac{e_{ind} - V_B}{R}$$, the current reverses

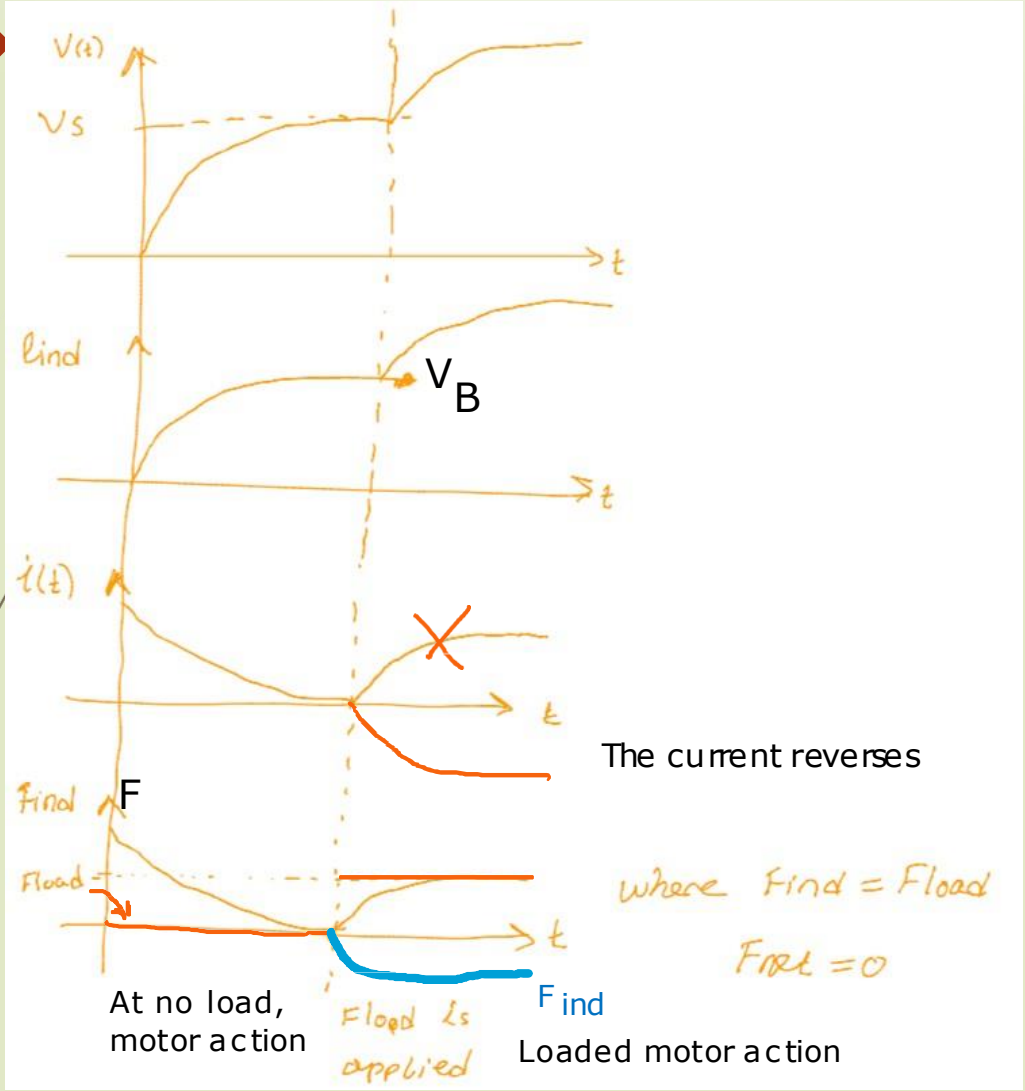
- Since this current now flows up through the bar, it induces a force in the bar given by

$$F_{ind} = ilB \qquad \text{to the left}$$

Summarize of a dc generator operation

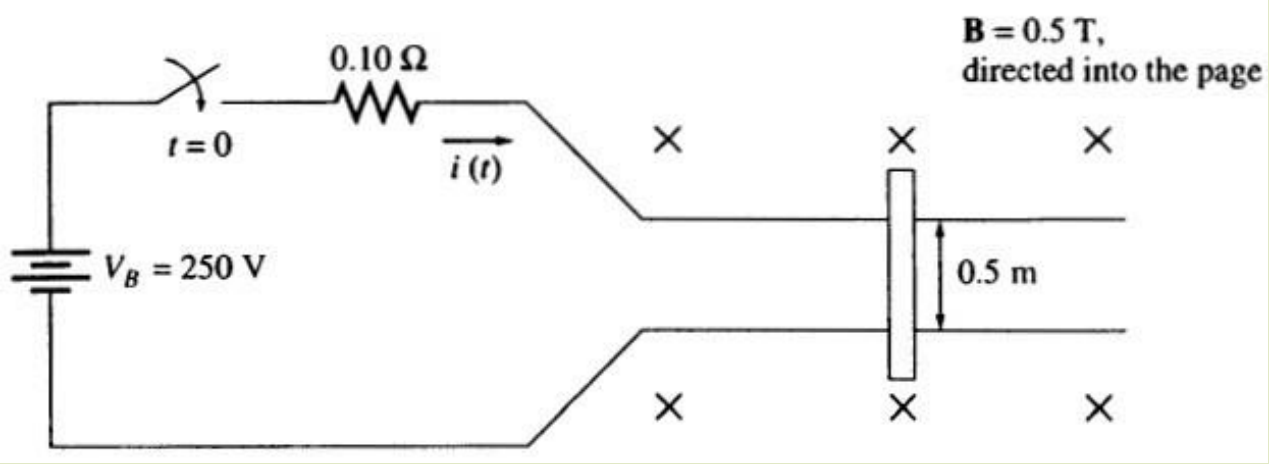
1. A force $\mathbf{F}_{app}$ is applied in the direction of motion; $\mathbf{F}_{net}$ is in the direction of motion.
2. Acceleration $a = F_{net}/m$ is positive, so the bar speeds up ($v\uparrow$).
3. The voltage $e_{ind} = v\uparrow Bl$ increases, and so $i = (e_{ind}\uparrow - V_B)/R$ increases.
4. The induced force $F_{ind} = i\uparrow lB$ increases until $|\mathbf{F}_{ind}| = |\mathbf{F}_{load}|$ at a higher speed v.
5. An amount of mechanical power equal to $F_{ind}v$ is now being converted to electric power $e_{ind}i$, and the machine is acting as a generator.

Graphical plots for generator action



Starting problems with dc linear machine

At starting conditions, the speed of the bar is zero, so $e_{ind} = 0$.



Thus, the current flow at starting is

$$i_{start} = \frac{V_B}{R} = \frac{250\ V}{0.1\ \Omega} = 2500\ \text{A}$$

Such currents can cause severe damage to a motor.

Starting problems with dc linear machine

Inserting an extra resistance into the circuit during starting to limit the current flow until e_{ind} builds up enough to limit it.

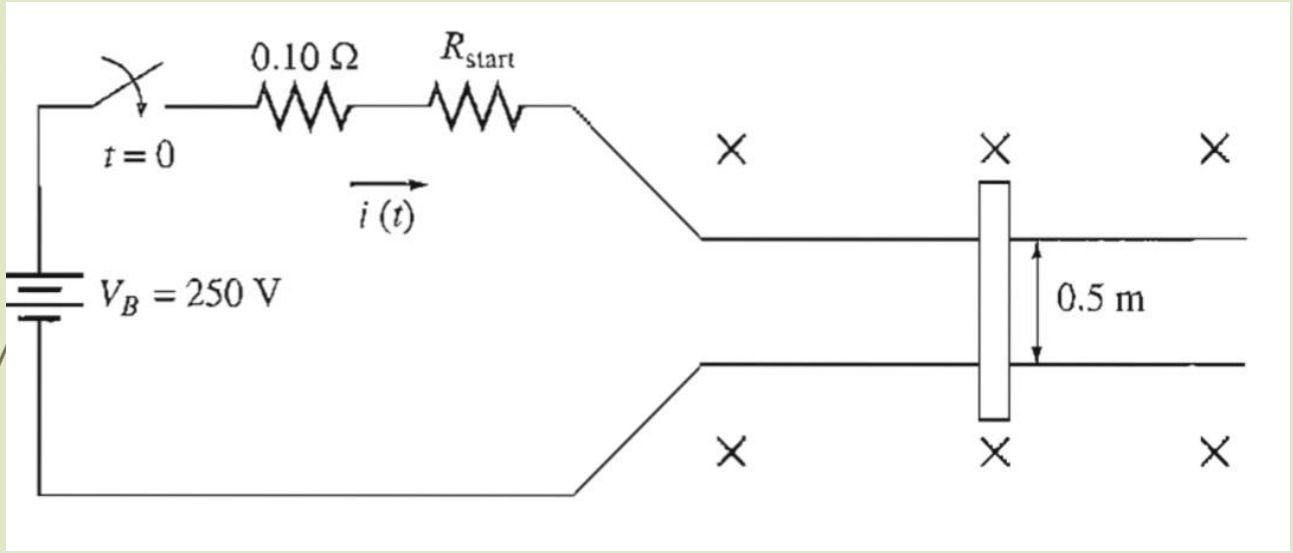


Fig 1-26 A linear dc machine with an extra series resistor inserted to control the starting current.

Example 1-10

Example 1–10. The linear dc machine shown in Figure 1–27a has a battery voltage of 120 V, an internal resistance of 0.3 Ω, and a magnetic flux density of 0.1 T.

(a) What is this machine's maximum starting current? What is its steady-state velocity at no load?

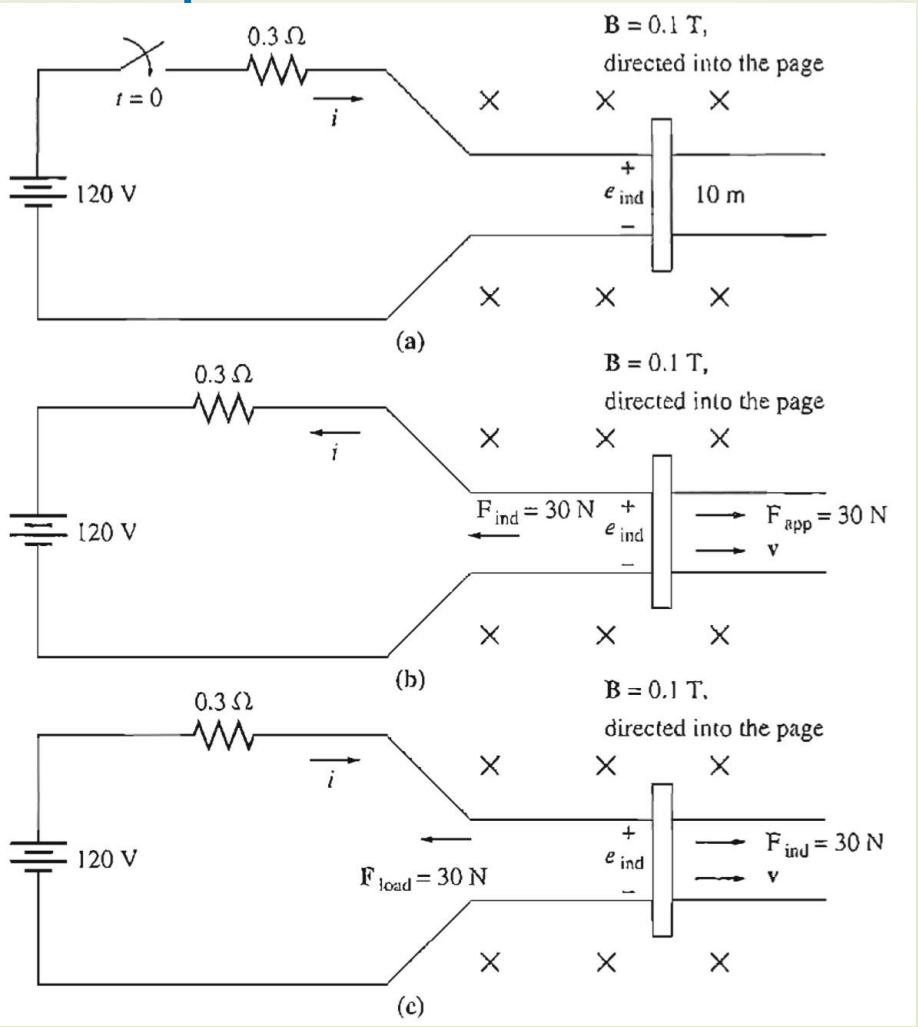
(b) Suppose that a 30-N force pointing to the right were applied to the bar. What would the steady-state speed be? How much power would the bar be producing or consuming? How much power would the battery be producing or consuming? Explain the difference between these two figures. Is this machine acting as a motor or as a generator?

(c) Now suppose a 30-N force pointing to the left were applied to the bar. What would the new steady-state speed be? Is this machine a motor or a generator now?

(d) Assume that a force pointing to the left is applied to the bar. Calculate speed of the bar as a function of the force for values from 0 N to 50 N in 10-N steps. Plot the velocity of the bar versus the applied force.

(e) Assume that the bar is unloaded and that it suddenly runs into a region where the magnetic field is weakened to 0.08 T. How fast will the bar go now?

Example 1-10



$i = \frac{250}{0.1} = 2500\ A$

10 times the rated current of the m/c

Fig. 1-27 The linear dc machnine of Example 1-10, (a) Starting conditions; (b) operating as a generator; (c) operating as a motor

Example 1-10

Solution

(a) At starting conditions, the velocity of the bar is 0, so $e_{ind} = 0$. Therefore,

$$i = \frac{V_B - e_{ind}}{R} = \frac{120\ \text{V} - 0\ \text{V}}{0.3\ \Omega} = 400\ \text{A}$$

When the machine reaches steady state, $\mathbf{F}_{ind} = 0$ and $i = 0$. Therefore,

$$VB = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl}$$

$$= \frac{120\ \text{V}}{(0.1\ \text{T})(10\ \text{m})} = 120\ \text{m/s}$$

Example 1-10

(b) Refer to Figure 1–27b. If a 30-N force to the right is applied to the bar, the final steady state will occur when the induced force $\mathbf{F}_{ind}$ is equal and opposite to the applied force $\mathbf{F}_{app}$, so that the net force on the bar is zero:

$$F_{app} = F_{ind} = ilB$$ generator action

Therefore,

$$i = \frac{F_{ind}}{lB} = \frac{30\ \text{N}}{(10\text{m})(0.1\ \text{T})}$$

$$= 30\ \text{A} \quad \text{flowing up through the bar}$$

The induced voltage e_{ind} on the bar must be

$$e_{ind} = V_B + iR$$

$$= 120\ \text{V} + (30\text{A})(0.3\ \Omega) = 129\ \text{V}$$

and the final steady-state speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{129\ \text{V}}{(0.1\ \text{T})(10\ \text{m})} = 129\ \text{m/s}$$

The bar is *producing* $P = (129\ \text{V})(30\ \text{A}) = 3870\ \text{W}$ of power, and the battery is *consuming* $P = (120\ \text{V})(30\ \text{A}) = 3600\ \text{W}$. The difference between these two numbers is the 270 W of losses in the resistor. This machine is acting as a *generator.*

Example 1-10

(c) Refer to Figure 1–25c. This time, the force is applied to the left, and the induced force is to the right. At steady state,

motor action

$$F_{app} = F_{ind} = ilB$$

$$i = \frac{F_{ind}}{lB} = \frac{30\ \text{N}}{(10\ \text{m})(0.1\ \text{T})}$$

$$= 30\ \text{A} \quad \text{flowing down through the bar}$$

The induced voltage e_{ind} on the bar must be

$$e_{ind} = V_B - iR$$

$$= 120\ \text{V} - (30\ \text{A})(0.3\ \Omega) = 111\ \text{V}$$

and the final speed must be

$$v_{ss} = \frac{e_{ind}}{Bl}$$

$$= \frac{111\ \text{V}}{(0.1\ \text{T})(10\ \text{m})} = 111\ \text{m/s}$$

This machine is now acting as a *motor*, converting electric energy from the battery into mechanical energy of motion on the bar.

Example 1-10

(e) If the bar is initially unloaded, then $e_{ind} = V_B$. If the bar suddenly hits a region of weaker magnetic field, a transient will occur. Once the transient is over, though, e_{ind} will again equal V_B.

This fact can be used to determine the final speed of the bar. The *initial speed* was 120 m/s. The *final speed* is

$$VB = e_{ind} = v_{ss}Bl$$

$$v_{ss} = \frac{V_B}{Bl}$$

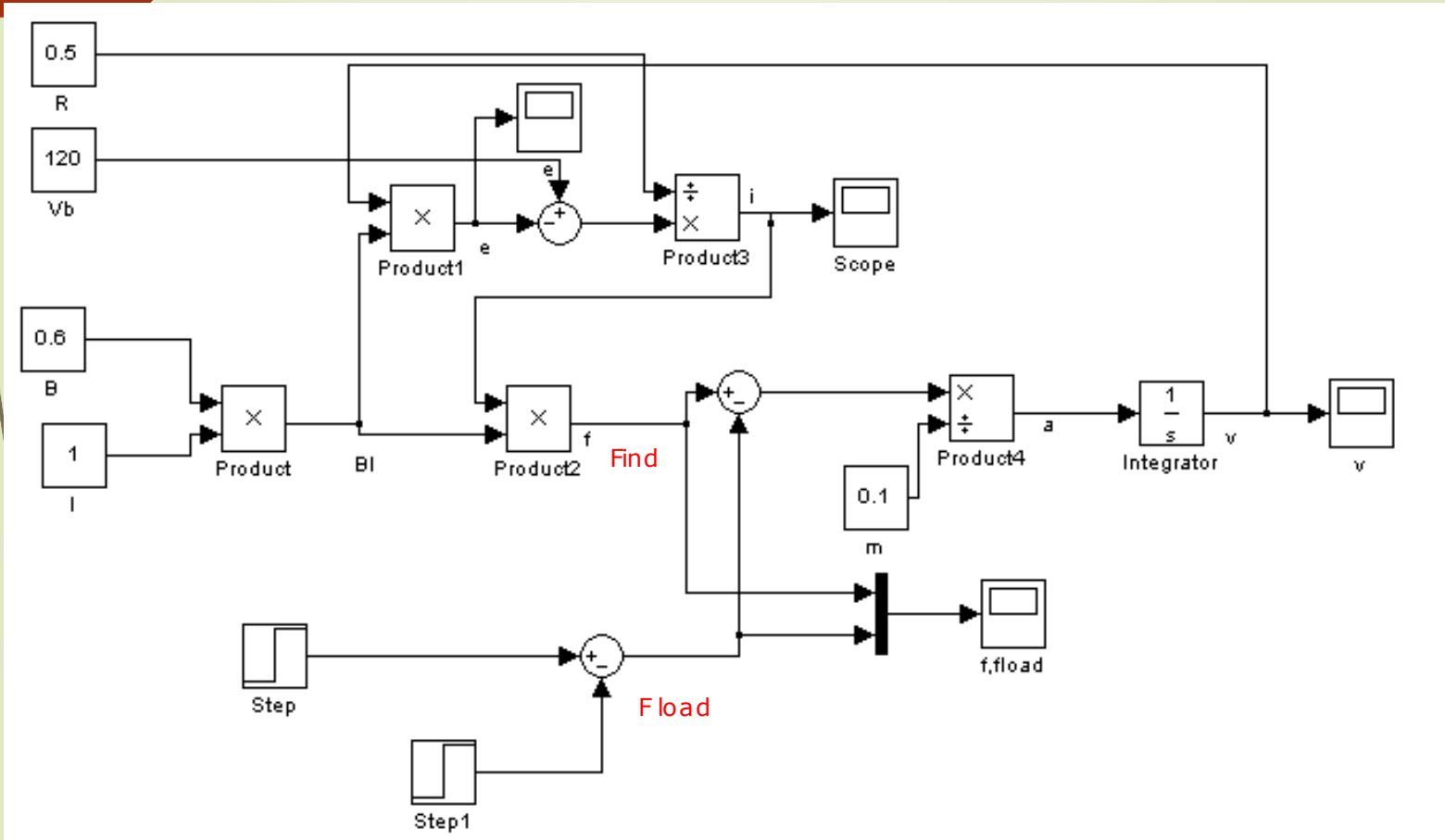
$$= \frac{120\ \text{V}}{(0.08\ \text{T})(10\ \text{m})} = 150\ \text{m/s}$$

Thus, when the flux in the linear motor weakens, the bar speeds up. The same behavior occurs in real dc motors: When the field flux of a dc motor weakens, it turns faster. Here, again, the linear machine behaves in much the same way as a real dc motor.

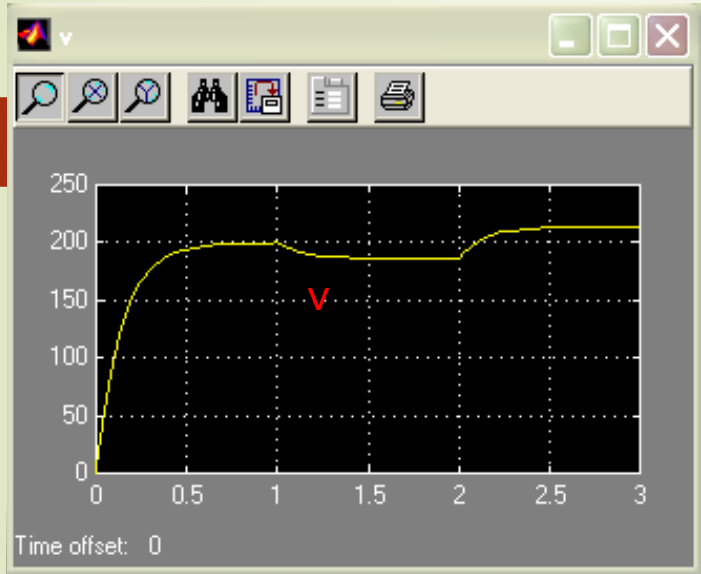
Linear dc motor simulation

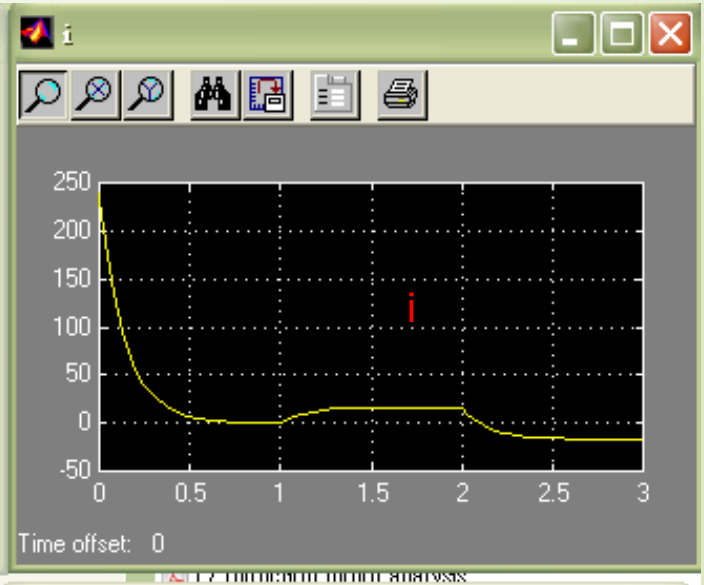
- **Equations:** (**only magnitudes**)
  - **F** = ilB, l: bar length
  - **e** = vBl
  - **i** = (Vb - e) / R
  - **dv/dt** = a = (F - Fload)/m
- **Simulation parameters:**
  - **Vb** = 120V, R = 0.3Ω, l = 1m
  - **B** = 0.6T, m = 0.1kg
  - **Fload** = 10(u-1) - 20(u-2) N

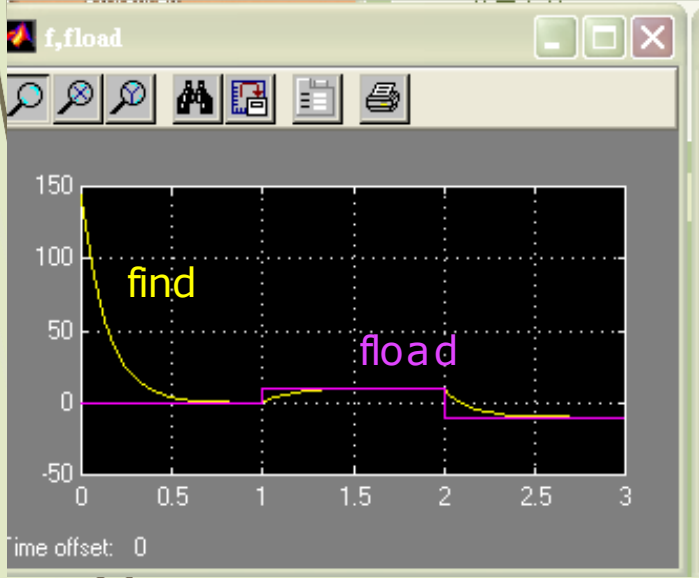
Linear dc motor simulation

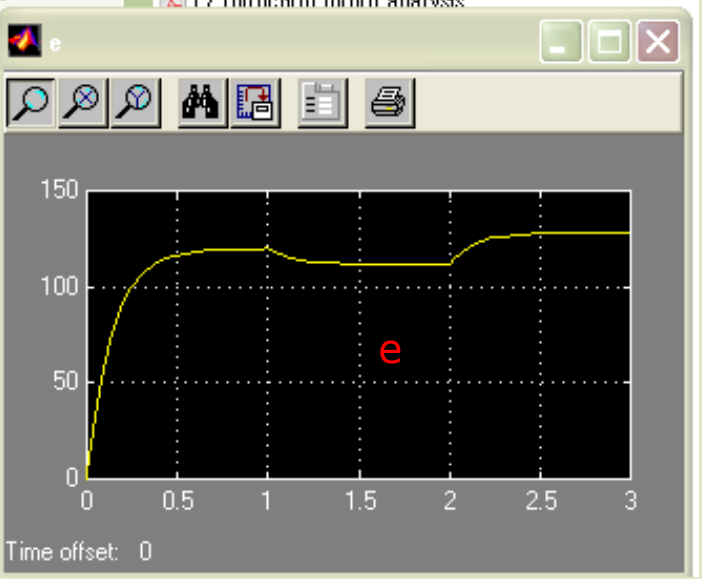


e=Blv, (Vb-e)/R=I, Bli=f, f-(fstep_t1- fstep_t2)/m=a, a/s=v

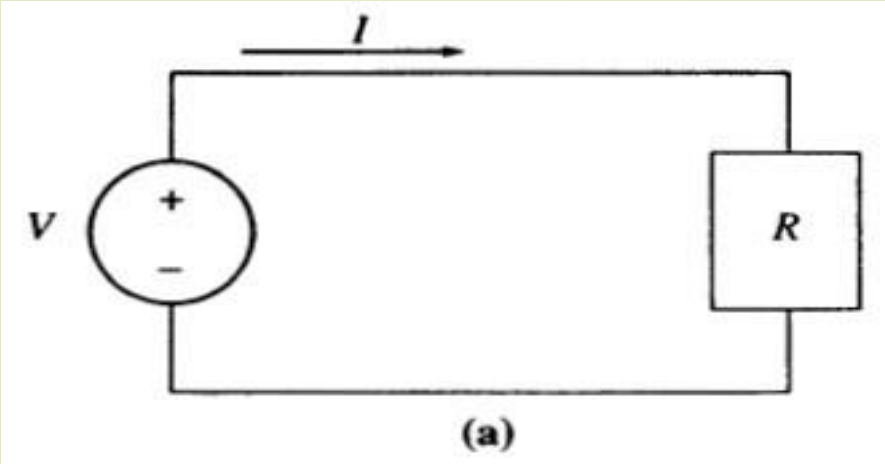

v
250
200
150
100
50
0
0
0.5
1
1.5
2
2.5
3
v
Time offset: 0


i
250
200
150
100
50
0
-50
0
0.5
1
1.5
2
2.5
3
i
Time offset: 0


f,fload
150
100
50
0
-50
0
0.5
1
1.5
2
2.5
3
find
fload
ime offset: 0


e
150
100
50
0
0
0.5
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1.5
2
2.5
3
e
Time offset: 0

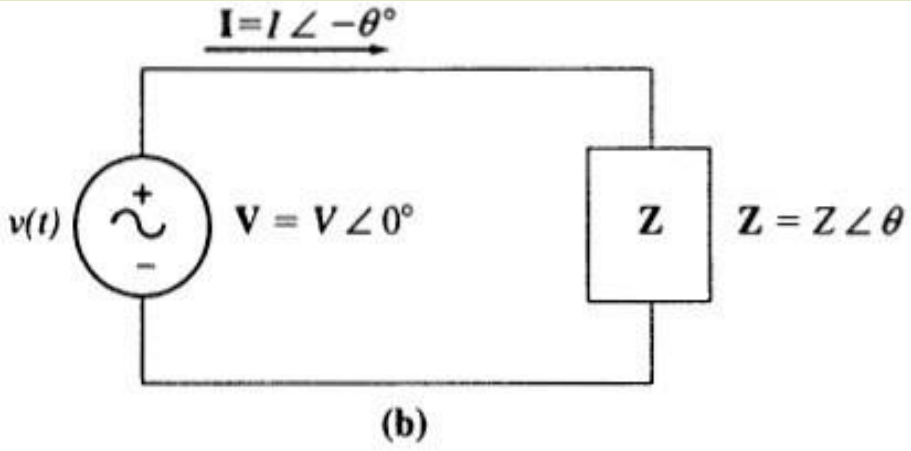
Real, reactive, and apparent power in AC circuits



$$P = VI$$

Real, reactive and apparent power in AC circuits

- AC source applies power to an impedance Z



$$v(t) = \sqrt{2}V\cos\omega t$$

$$i(t) = \sqrt{2}I\cos(\omega t - \theta)$$

Instantaneous power

The instantaneous power supplied to this load at any time t is

$$p(t) = v(t)i(t) = 2VI \cos \omega t \cos(\omega t - \theta) \qquad (1\text{–}58)$$

The angle θ in this equation is the *impedance angle* of the load. For inductive loads, the impedance angle is positive, and the current waveform lags the voltage waveform by θ degrees.

If we apply trigonometric identities to Equation (1–58), it can be manipulated into an expression of the form

$$p(t) = VI \cos \theta (1 + \cos 2\omega t) + VI \sin \theta \sin 2\omega t \qquad (1\text{–}59)$$

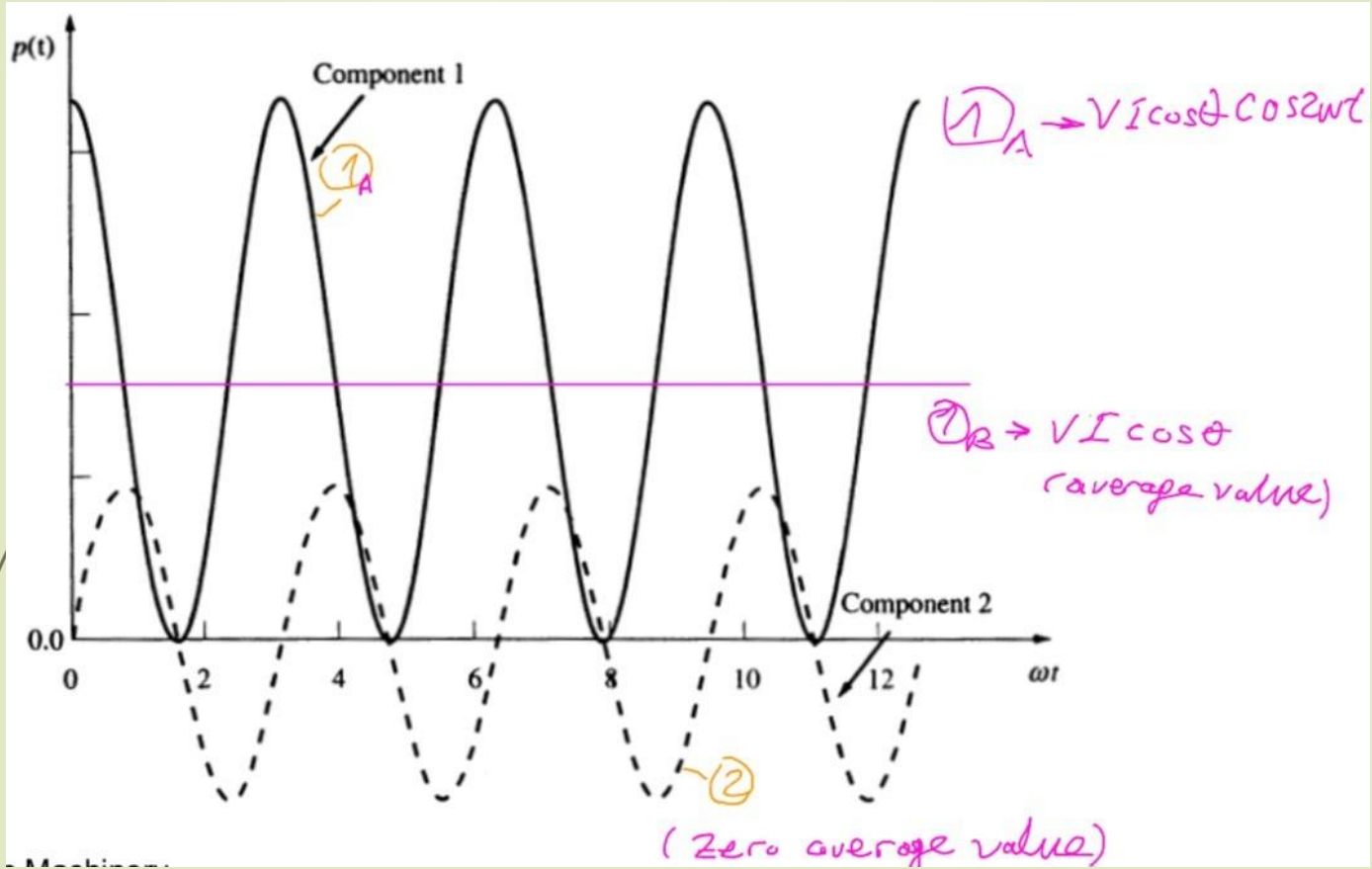
(1) (2)

where $\cos(\omega t - \theta) = \cos\omega t \cos\theta + \sin\omega t \sin\theta$

$p(t) = 2VI(\cos^2\omega t \cos\theta + \cos\omega t \sin\omega t \sin\theta)$

where $\cos^2\omega t = \frac{1}{2}[1 + \cos(2\omega t)]$ and $\cos\omega t \sin\omega t = \frac{1}{2}\sin 2\omega t$

Instantaneous power



Average power and reactive power

Note that the *first* term of the instantaneous power expression is always positive, but it produces pulses of power instead of a constant value. The average value of this term is

$$P = VI \cos \theta \qquad (1\text{–}60)$$

which is the *average* or *real* power (P) supplied to the load by term 1 of the Equation (1–59). The units of real power are watts (W), where 1 W = 1 V × 1 A.

Note that the *second* term of the instantaneous power expression is positive half of the time and negative half of the time, so that *the average power supplied by this term is zero.* This term represents power that is first transferred from the source to the load, and then returned from the load to the source. The power that continually bounces back and forth between the source and the load is known as *reactive power* (Q). Reactive power represents the energy that is first stored and then released in the magnetic field of an inductor, or in the electric field of a capacitor.

Reactive power Q and apparent power S

1. Reactive power Q (var) is defined from instantaneous power

$$Q = VI \sin \theta$$

2. Apparent power S (VA) is defined to represent the product of voltage and current magnitudes

$$S = VI$$

Alternative Forms of the Power Equations

If a load has a constant impedance, then Ohm's law can be used to derive alternative expressions for the real, reactive, and apparent powers supplied to the load. Since the magnitude of the voltage across the load is given by

$$V = IZ \tag{1–63}$$

substituting Equation (1–63) into Equations (1–60) to (1–62) produces equations for real, reactive, and apparent power expressed in terms of current and impedance:

$$P = I^2 Z \cos\theta \tag{1–64}$$

$$Q = I^2 Z \sin\theta \tag{1–65}$$

$$S = I^2 Z \tag{1–66}$$





where $|Z|$ is the magnitude of the load impedance Z.

Since the impedance of the load Z can be expressed as

$$Z = R + jX = |Z|\cos\theta + j|Z|\sin\theta$$

we see from this equation that $R = |Z|\cos\theta$ and $X = |Z|\sin\theta$, so the real and reactive powers of a load can also be expressed as

$$P = I^2 R = I^2 |Z| \cos\theta \tag{1–67}$$

$$Q = I^2 X = I^2 |Z| \sin\theta \tag{1–68}$$

where R is the resistance and X is the reactance of load Z.

Power in AC circuit

$i(t) = (I/\mathrm{sqr}(2))\cos(wt-\theta)$ rms

The chosen *v* and *i* include all possibilities because

- If the load is purely resistive: $\theta = 0°$
- If the load is purely inductive: $\theta = 90°$
- If the load is purely capacitive: $\theta = -90°$

For a network that is primarily inductive, θ is positive (*v* leads *i* or *i* lags *v*)

For a network that is primarily capacitive, θ is negative (*i* leads *v*)

Power in AC circuit

Positive power means that power has been distributed from supply into circuit.

Negative power means that the power has been distributed from circuit into supply.

There are three type of power in AC circuit:

i) Average power, P
ii) Apparent Power, S
iii) Reactive Power, Q

Complex power representation

For simplicity in computer calculations, real and reactive power are sometimes represented together as a *complex power* **S**, where

$$\mathbf{S} = P + jQ \tag{1–69}$$

The complex power **S** supplied to a load can be calculated from the equation

$$\mathbf{S} = \mathbf{VI}^* = V_{rms} I_{rms}^* \tag{1–70}$$

where the asterisk represents the complex conjugate operator.

where $|S| = V_{rms} I_{rms} = \sqrt{P^2 + Q^2}$

Complex power representation

To understand this equation, let's suppose that the voltage applied to a load is $\mathbf{V} = V \angle \alpha$ and the current through the load is $\mathbf{I} = I \angle \beta$. Then the complex power supplied to the load is

$$\mathbf{S} = \mathbf{VI}^* = (V\angle\alpha)(I\angle -\beta) = VI\angle(\alpha - \beta) = VI\cos(\alpha - \beta) + jVI\sin(\alpha - \beta)$$

$= VI\, e^{j(\alpha-\beta)}$, with $\theta = \alpha - \beta$

The impedance angle θ is the difference between the angle of the voltage and the angle of the current ($\theta = \alpha - \beta$), so this equation reduces to

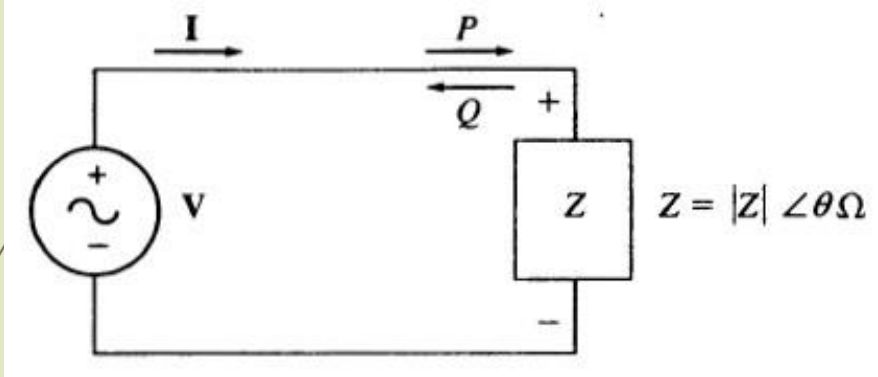
$$\mathbf{S} = VI\cos\theta + jVI\sin\theta = P + jQ$$

Power direction

The direction of Q

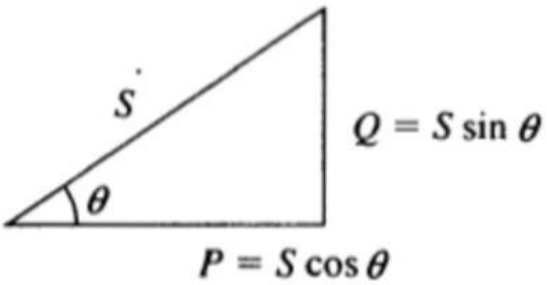
+ve for the first cycle

-ve for the second cycle.



$$\mathbf{I} = \frac{\mathbf{V}}{Z} = \frac{V\angle 0^\circ}{|Z|\angle\theta} = \frac{V}{|Z|}\angle -\theta$$

Power factor



$$\cos\theta = \frac{P}{S}$$

$$\sin\theta = \frac{Q}{S}$$

$$\tan\theta = \frac{Q}{P}$$

The quantity cos θ is usually known as the *power factor* of a load. The power factor is defined as the fraction of the apparent power S that is actually supplying real power to a load. Thus,

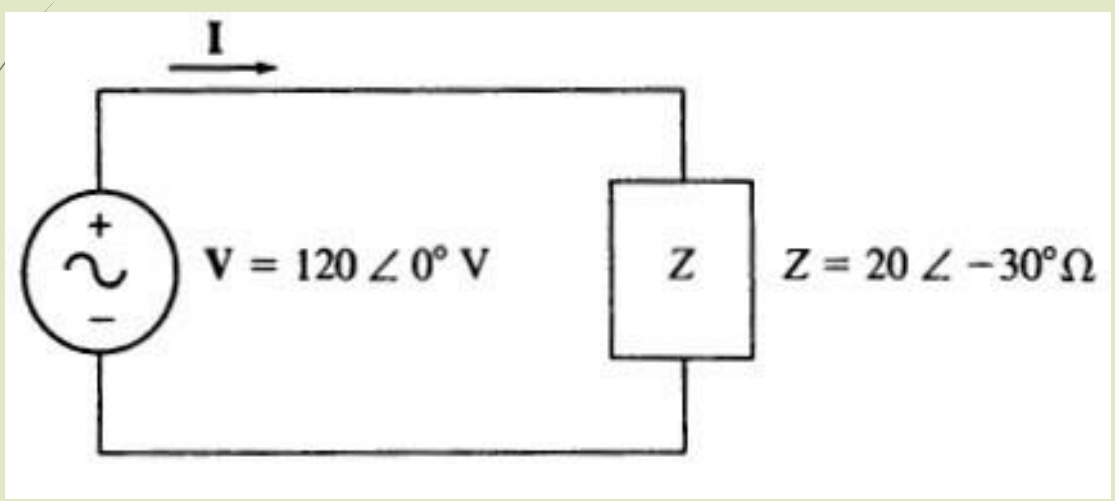
$$\text{PF} = \cos\theta = \frac{P}{|S|} \qquad (1\text{–}71)$$

where θ is the impedance angle of the load.

where $|S| = VI$

Example 1-11

Example 1–11. Figure 1–34 shows an ac voltage source supplying power to a load with impedance $Z = 20\angle -30^\circ\ \Omega$. Calculate the current **I** supplied to the load, the power factor of the load, and the real, reactive, apparent, and complex power supplied to the load.



Example 1-11

Solution

The current supplied to this load is

$$\mathbf{I} = \frac{\mathbf{V}}{Z} = \frac{120\angle 0^\circ \text{ V}}{20\angle -30^\circ\ \Omega} = 6\angle 30^\circ \text{ A}$$

The power factor of the load is

$$\text{PF} = \cos\theta = \cos(-30^\circ) = 0.866 \text{ leading} \qquad (1\text{–}71)$$

(Note that this is a capacitive load, so the impedance angle θ is negative, and the current *leads* the voltage.)

The real power supplied to the load is

$$P = VI\cos\theta \qquad (1\text{–}60)$$

$$P = (120 \text{ V})(6 \text{ A})\cos(-30^\circ) = 623.5 \text{ W}$$

The reactive power supplied to the load is

$$Q = VI\sin\theta \qquad (1\text{–}61)$$

$$Q = (120 \text{ V})(6 \text{ A})\sin(-30^\circ) = -360 \text{ VAR}$$

Example 1-11

The apparent power supplied to the load is

$$S = VI \tag{1–62}$$

$$Q = (120\ \text{V})(6\ \text{A}) = 720\ \text{VA}$$

The complex power supplied to the load is

$$\mathbf{S} = \mathbf{VI}^* \tag{1–70}$$

$$= (120\angle 0°\ \text{V})(6\angle -30°\ \text{A})^*$$

$$= (120\angle 0°\ \text{V})(6\angle 30°\ \text{A}) = 720\angle 30°\ \text{VA}$$

$$= 623.5 - j360\ \text{VA}$$

Notes about total P, Q, and S

- First, the total apparent power, S_T can be determined from the total average P_T and total reactive powers Q_T and **cannot** be determined from the apparent powers of each branch.
- Second, and more important, it is **not necessary** to consider the series-parallel arrangement of branches. In other words, the total real P_T, total reactive Q_T, or total apparent power S_T is independent of whether the loads are in series, parallel, or series-parallel.



θ_1 : positive ⇓ Inductive load



θ_2 : negative ⇓ Capacitive load



$$jQ = jQ_L - jQ_C$$

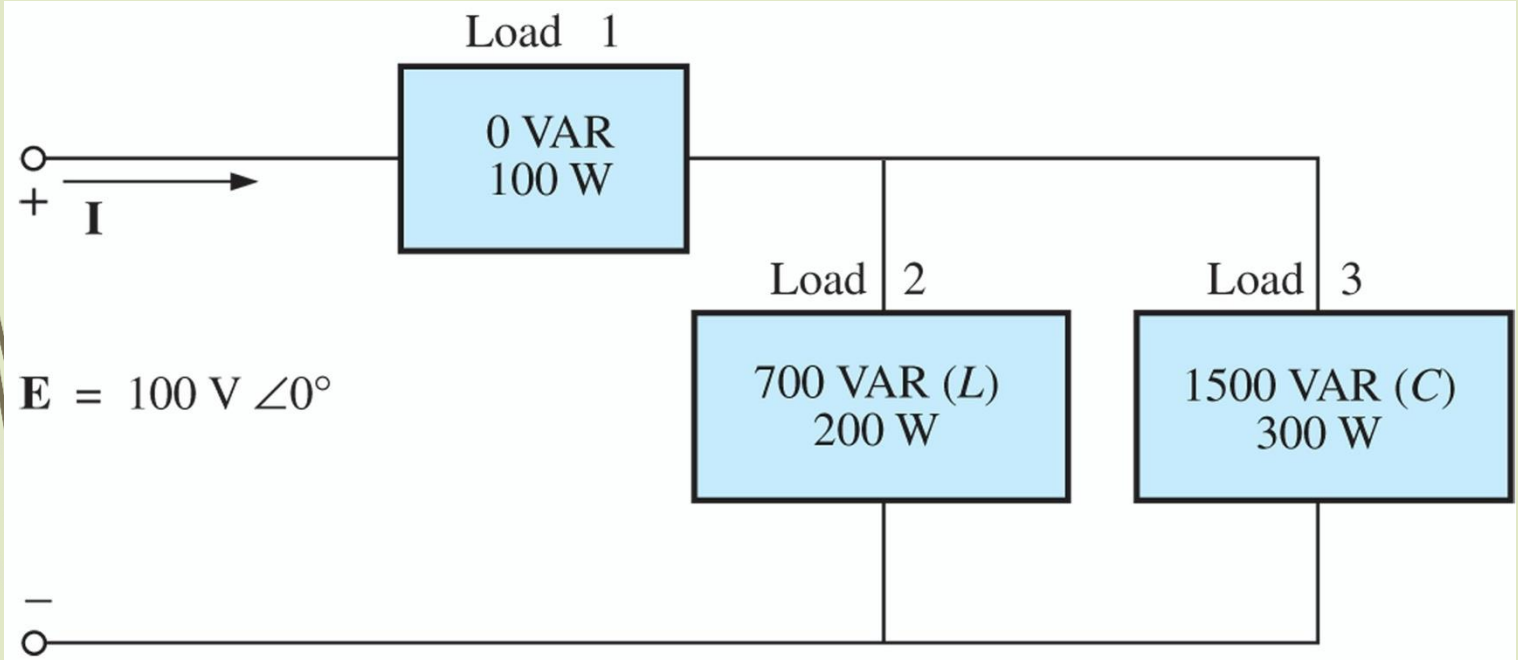
$$S = P + jQ$$

$$\theta_1 - \theta_2 = \theta$$

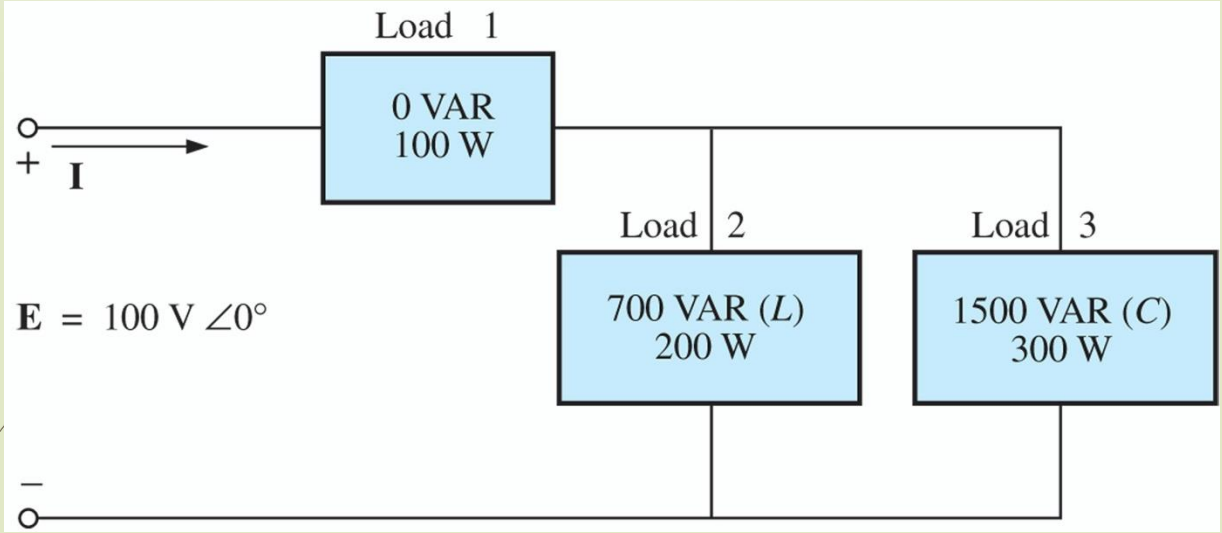


Example

1. Find the total real power P_T, total reactive power Q_T, and volt-amperes S_T and draw the power triangle.
2. Find the power factor (PF)
3. Find the current in phasor form.



Solution



1. Find total power

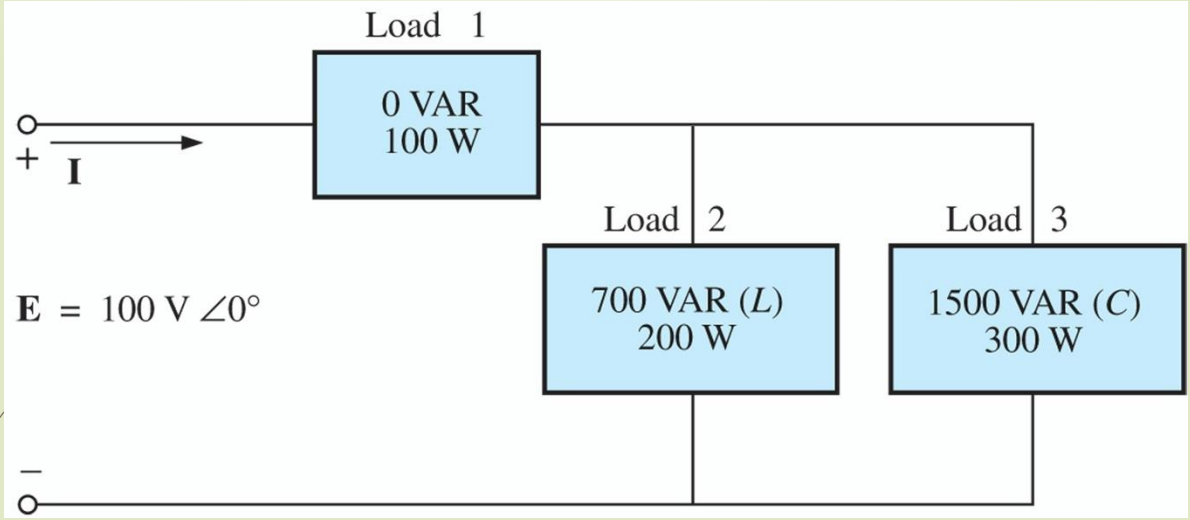
Load 1

$$S = \sqrt{P^2 + Q^2}$$

$$P_1 = 100 \text{ W};$$
$$Q_1 = 0 \text{ VAR}$$

$$S_1 = 100 + j0$$
$$= 100 \text{ VA}$$

Solution (cont'd)



Load 2

$$S = \sqrt{P^2 + Q^2}$$

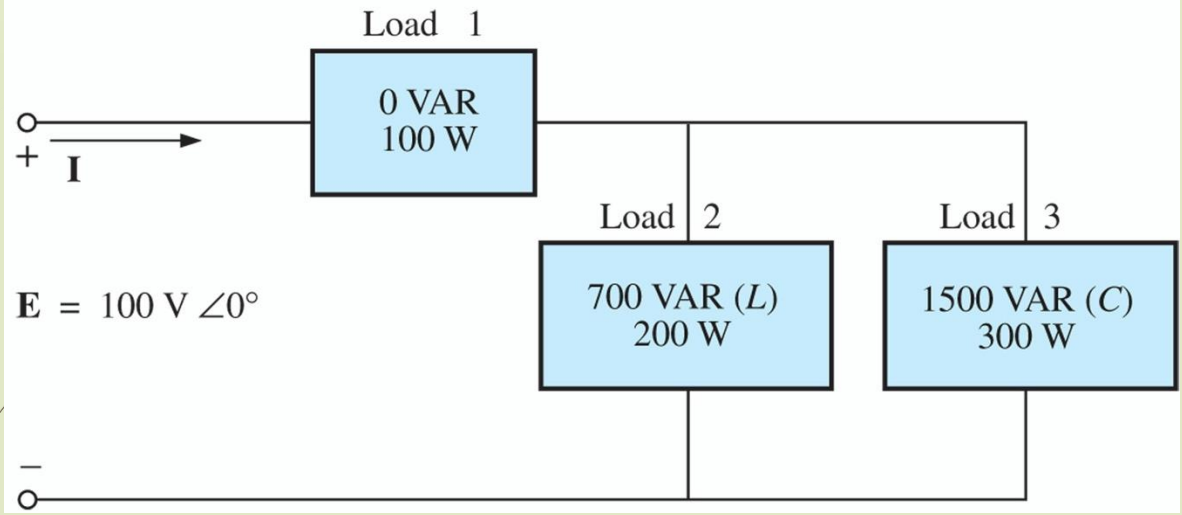
$P_2 = 200\ \text{W};$

$Q_2 = 700\ \text{VAR}$

$S_2 = 200 + j700$

$= 728\ \text{VA}$

Solution (cont'd)



Load 3

$$S = \sqrt{P^2 + Q^2}$$

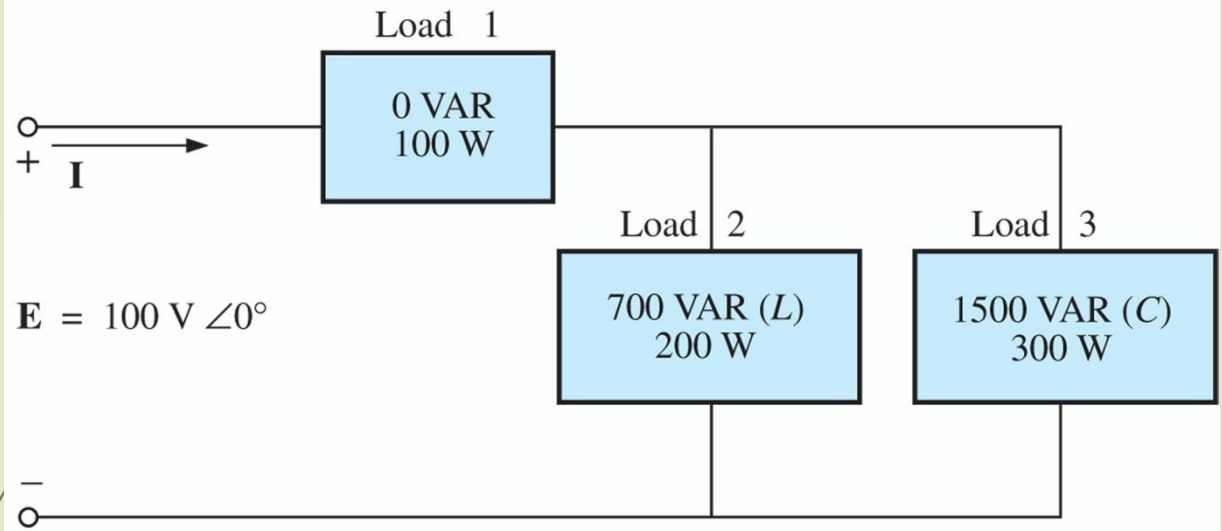
$P_3 = 300\ \text{W};$

$Q_3 = 1500\ \text{VAR}$

$S_3 = 300 - j1500$

$= 1529.71\ \text{VA}$

Solution (cont'd)



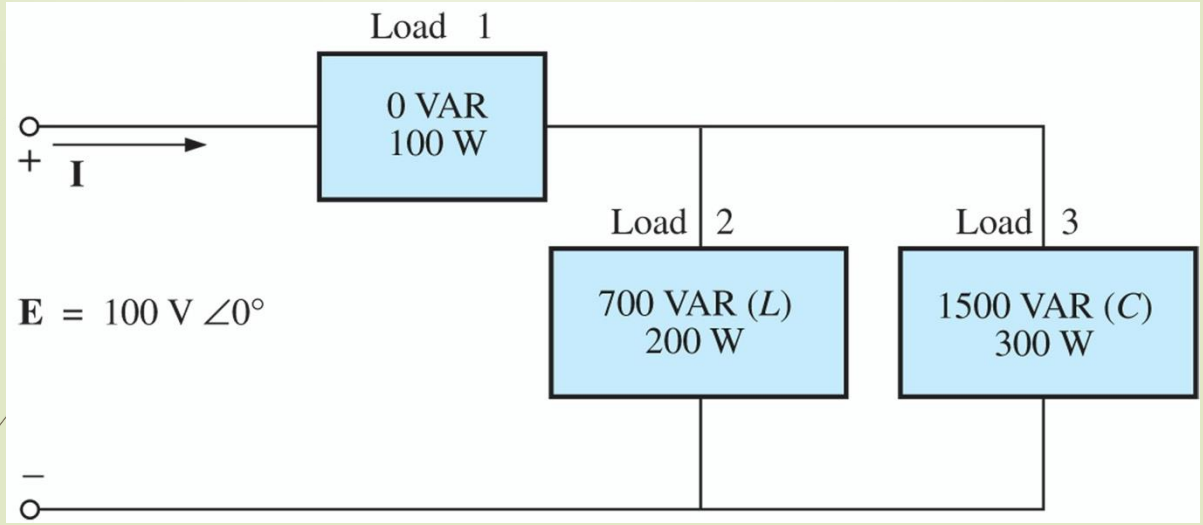
Total real power =>

$$P_T = P_1 + P_2 + P_3$$

$$= 100 + 200 + 300$$

$$P_T = 600\ \text{W}$$

Solution (cont'd)



Total reactive power =>

$$Q_T = Q_1 + Q_2 + Q_3$$
$$= 0 + 700 - 1500$$
$$= -800 \text{ VAR}$$

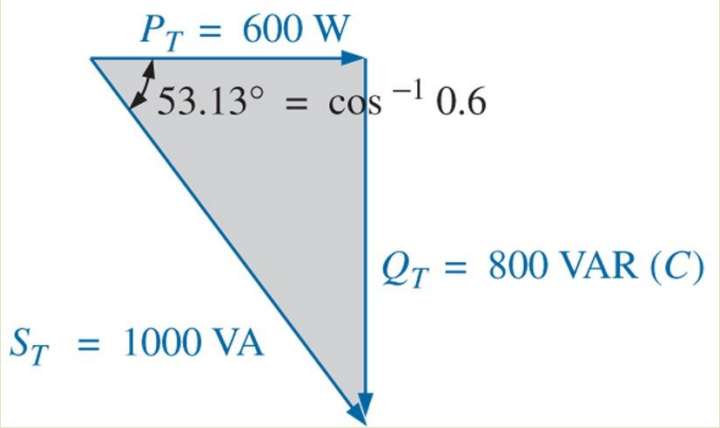
$$Q_T = 800 \text{ VAR(C)}$$

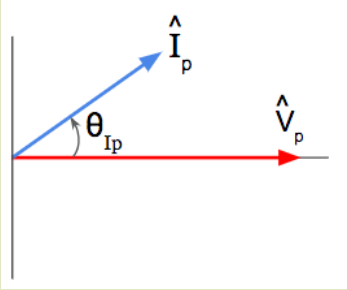
Solution (cont'd)

2. The power factor (PF)

$$PF = \frac{P_T}{S_T} = \frac{600}{1000}$$

$PF = 0.6$ *leading*





$\hat{S} = |S|\angle\theta \quad \hat{I}_p = |I_p|\angle\theta_{Ip} \quad \hat{V}_p = |V_p|\angle 0°$

$\theta = -\theta_{Ip}$

$\theta_{Ip} = -\theta$

Only when: $\theta_{Vp} = 0°$

Solution (cont'd)

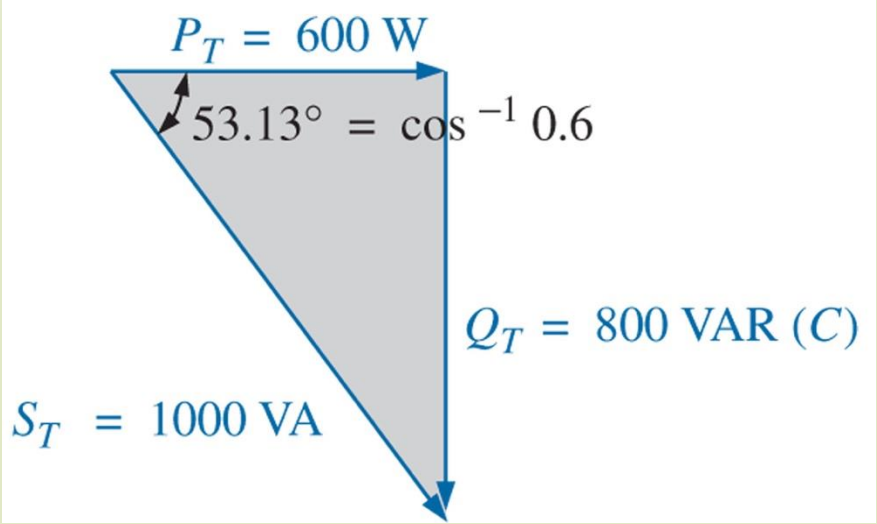
3. The current

$$S_T = 1000\ \text{VA}$$

$$VI = 1000\ \text{VA}$$

$$I = \frac{1000\ \text{VA}}{100\ \text{V}}$$

$$I = 10\ \text{A}$$



Since *PF* is **leading,** **I** leads **E**, predominantly **capacitive** circuit.

$$\mathbf{I} = 10\ \text{A} \angle +53.13°$$

Type of Power	Formulas	Units
Average Power	$P = V_{rms} I_{rms} \cos\theta$ $= I_{rms}^2 R$ $= \frac{V_{rms}^2}{R}$	Watt
Reactance Power	$Q = V_{rms} I_{rms} \sin\theta$ $= I_{rms}^2 X$ $= \frac{V_{rms}^2}{X}$	VAR
Apparent Power	$S = V_{rms} I_{rms}$ $= I_{rms}^2 Z$ $= \sqrt{P^2 + Q^2}$	VA
Complex Power	$S_C = V_{rms} I_{rms}^*$ $= S\angle \pm\theta$ $= P \pm Q$	VA